Zeroth-Order Optimization

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Contents

Chapter 1

Zeroth-Order Mirror Descent

In this excerpt we prove that Zero-Order Mirror Descent converges with high probability.

Assumption 1.1. *f is ρ-Lipschitz continuous and convex over a closed convex set $Q \subset \mathbb{R}^d$.*

Definition 1.2. $w : Q \to \mathbb{R}^d$ is a DGF if it is 1-strongly convex w.r.t. $\|\cdot\|$, in symbols, $w(y) \geq w(x) + \langle \nabla w(x), y - x\rangle + \frac{1}{2}\|y - x\|^2$. The Bregman divergence is defined

$$V_x(y) = w(y) - \langle \nabla w(y), y - x\rangle - w(x)$$

We have $V_x(x) = 0$ and $V_x(y) \geq \frac{1}{2}\|x - y\|^2 \geq 0$.

Classic examples are $w(x) = \frac{1}{2}\|x\|_2^2$ for the ℓ^2 norm which gives $V_x(y) = \frac{1}{2}\|y - x\|_2^2$, and $w(y) = \sum_i y_i \log(y_i)$ for the ℓ^1 norm with $Q = \Delta$, which gives KL divergence $V_x(y) = \sum_i y_i \log(y_i/x_i) \geq \frac{1}{2}\|x - y\|_1^2$.

1.1 Classic Mirror Descent

Definition 1.3 (Mirror Descent Step)**.** The mirror descent step with step length α is

$$x^+ = \mathrm{Mirr}_x(\alpha \nabla f(x)) \quad \text{where} \quad \mathrm{Mirr}_x(\xi) = \operatorname*{argmin}_{y \in Q} \{V_x(y) + \langle \xi, y - x\rangle\}.$$

Mirror descents core lemma is this:

Lemma 1.4. *If $x_{t+1} = \mathrm{Mirr}_{x_t}(\alpha \nabla f(x_k))$, then*

$$\forall u \in Q, \quad \alpha(f(x_t) - f(u)) \leq \alpha\langle \nabla f(x_t), x_t - u\rangle \leq \frac{\alpha^2}{2}\|\nabla f(x_t)\| + V_{x_t}(u) - V_{x_{t+1}}(u).$$

Proof. Notice

$$\alpha\langle \nabla f(x_t), x_t - u\rangle = \langle \alpha \nabla f(x_t), x_t - x_{t+1}\rangle + \langle \alpha \nabla f(x_t), x_{t+1} - u\rangle$$

Since Q is convex and $x_{t+1} = \mathrm{Mirr}_{x_t}(\alpha \nabla f(x))$ is minimal, it satisfies the first order condition

$$\langle \nabla V_{x_t}(x_{t+1}) + \alpha \nabla f(x_t), u - x_{t+1} \rangle \geq 0 \quad \forall u \in Q$$

Therefore the above is at most

$$\langle \alpha \nabla f(x_t), x_t - x_{t+1} \rangle + \langle -\nabla V_{x_k}(x_{t+1}), x_{t+1} - u \rangle$$

Using the three-point identity

$$\langle -\nabla V_x(y), y - u \rangle = V_x(u) - V_y(u) - V_x(y)$$

we get the above equals

$$\begin{aligned}
&\langle \alpha \nabla f(x_t), x_t - x_{t+1} \rangle + V_{x_t}(u) - V_{x_{t+1}}(u) - V_{x_t}(x_{t+1}) \\
\leq &\left(\langle \alpha \nabla f(x_t), x_t - x_{t+1} \rangle - \frac{1}{2} \|x_t - x_{t+1}\|^2 \right) + (V_{x_t}(u) - V_{x_{t+1}}(u)) \\
\leq &\ \frac{\alpha^2}{2} \|\nabla f(x_t)\|^2 + (V_{x_t}(u) - V_{x_{t+1}}(u))
\end{aligned}$$

After recalling that $V_x(y) \geq \frac{1}{2}\|x - y\|^2$, using Cauchy-Schwarz $\langle a, b\rangle \leq \|a\| \cdot \|b\|$ and Young's inequality $2ab \leq a^2 + b^2$ in that order.

The first inequality follows because by convexity $f(u) \geq f(x_t) + \langle \nabla f(x_t), u - x_t \rangle$. □

Using this, we can prove Mirror Descent converges.

Lemma 1.5. *For $\bar{x} = \frac{1}{T}\sum_{t=0}^{T-1} f(x_t)$, if we choose $\alpha = \sqrt{2\Theta/\rho^2 T}$, we have*

$$f(\bar{x}) - f(x^*) \leq \rho \sqrt{\frac{2\Theta}{T}}$$

Proof. Telescoping, we get

$$\sum_{t=0}^{T-1} \alpha(f(x_t) - f(x^*)) \leq \sum_{t=0}^{T-1} \frac{\alpha^2}{2} \|\nabla f(x_t)\|^2 + V_{x_0}(x^*) - V_{x_T}(x^*)$$

Using convexity, $f(\bar{x}) \leq 1/T \sum_{t=0}^{T-1} f(x_t)$, and recalling that f is ρ-Lipschitz continuous, we have

$$f(\bar{x}) - f(x^*) \leq \frac{\alpha}{2}\rho^2 + \frac{1}{\alpha}\Theta,$$

where Θ is any number satisfying $\Theta \geq V_{x_0}(x^*)$. Now by choosing $\alpha = \sqrt{2\Theta/\rho^2 T}$ to balance the two terms, we are done. □

1.2 Zero-Order Mirror Descent

Throughout the rest of this excerpt the norm $\|\cdot\|$ is the ℓ^2 norm and $Q = \mathbb{R}^d$.

Now, notice that the only place in the proof of Lemma 1.4 where we used that the update is evaluated at the true gradient $\nabla f(x_k)$ was to upper bound $f(x_k) - f(u)$. The gradient was not used in the proof, actually. So we get this lemma for free:

Lemma 1.6. *If $x_{k+1} = \mathrm{Mirr}_{x_k}(\alpha g_k)$, then*

$$\forall u \in Q \quad \alpha\langle g_k, x_k - u\rangle \leq \frac{\alpha^2}{2}\|g_k\|^2 + V_{x_k}(u) - V_{x_{k+1}}(u).$$

Notice now that for any $u \in Q$,

$$\alpha(f(x_k) - f(u)) \leq \alpha\langle \nabla f(x_k), x_k - u\rangle = \alpha\langle \tilde{\nabla} f(x_k), x_k - u\rangle + \alpha\langle \nabla f(x_k) - \tilde{\nabla} f(x_k), x_k - u\rangle$$

Definition 1.7 (Zero-Order Estimator)**.** The Zero-Order gradient estimator is defined as follows

$$\tilde{\tilde{\nabla}} f(x) = \frac{f(x + \sigma p) - f(x - \sigma p)}{2\sigma} \cdot p$$

where $p \sim \mathcal{N}(0, I_d)$.

For simplicity of exposition in this chapter only we will assume there is no quadratic error and $\tilde{\nabla} f(x_t) = p_t p_t^\top \nabla f(x_t)$. Thus the above equals,

$$\begin{aligned} &\alpha\langle \nabla \tilde{f}(x_t), x_t - u\rangle + \alpha\langle (I - p_t p_t^\top)\nabla f(x_t), x_t - u\rangle \\ &\leq \frac{\alpha^2}{2}\left\|\tilde{\nabla} f(x_t)\right\|^2 + V_{x_t}(u) - V_{x_{t+1}}(u) + \alpha\langle (I - p_t p_t^\top)\nabla f(x_t), x_t - u\rangle \end{aligned} \tag{1.1}$$

This lemma explains how to deal with the first term.

Lemma 1.8. *With probability at least $1 - \delta$, we have for every $1 \leq t \leq T$,*

$$|p_t p_t^\top \nabla f(x_t)| \leq c\sqrt{d}\log(cT/\delta)\|\nabla f(x_t)\|$$

Proof. See Theorem 6 from EGGROLL convergence. □

The main idea behind the rest of this proof is that the mean-0 martingale error

$$\sum_{t=0}^{T-1} \alpha\langle (I - p_k p_k^\top)\nabla f(x_k), x_k - u\rangle$$

can hopefully be bounded by the other terms of the equation. In this case, it ends up being the "energy" term $V_{x_t}(u)$ and the Lipschitz constant ρ.

The main lemma which is different from the normal mirror descent bounds is the following.

Lemma 1.9. *Let $\{\mathcal{F}_t\}$ be a filtration. Assume that $g_t, d_t \in \mathcal{F}_{t-1}$, and assume $p_t \sim \mathcal{N}(0, I)$ with $p_t \in \mathcal{F}_t$, $p_t \perp \mathcal{F}_{t-1}$. Define*

$$E_\tau = \sum_{t=0}^{\tau-1} g_t^\top (I - p_t p_t^\top) d_t$$

Then for any θ, $0 < \delta < 1/e$, with probability at least $1 - \delta$,

$$|E_\tau| \leq c\theta \sum_{t=0}^{\tau-1} \log\Big(\frac{c\tau}{\delta}\Big) \|g_t\|^2 \|d_t\|^2 + \frac{1}{\theta} \log\Big(\frac{c\tau}{\delta}\Big).$$

Proof. For each t let

$$A_t = \left\{ L_\delta \leq |p_t^\top d_t| \leq \sqrt{3 \log(c/\delta) \|d_t\|^2} \right\}, \quad \mathbb{P}_{\mathcal{F}_{t-1}}(A_t) \geq 1 - \delta$$

Then by a lemma from before we know that $\mathbb{E}_{\mathcal{F}_{t-1}}\big[(I - p_t p_t^\top) d_t; A_t\big] = 0$. In particular,

$$\mathbb{E}_{\mathcal{F}_{t-1}}\Big[g_t^\top (I - p_t p_t^\top) d_t; A_t\Big] = 0.$$

Dropping the subscript t for simplicity,

$$\mathbb{E}\left[\exp\left(\frac{(g^\top p p^\top d \mathbb{1}_A)^2}{t^2}\right)\right] \leq \mathbb{E}\left[\exp\left(\frac{3\log(C/\delta) \|d_t\|^2 (p^\top g)^2}{t^2}\right)\right]$$

Now, $p^\top g$ is $\mathcal{N}(0, \|g\|^2)$ random variable. Therefore, $\mathbb{E}\Big[\exp\Big((p^\top g)^2 / 4\|g\|^2\Big)\Big] \leq 2$. Now, taking $c \geq 1$ and $0 < \delta \leq 1/e$, we have

$$(g^\top (I - pp^\top) d)^2 \leq 2(g^\top d)^2 + 2(g^\top p p^\top d)^2$$

So, the random variable $Q_t = g_t^\top (I - p_t p_t^\top) d_t \mathbb{1}_{A_t}$ is $24 \|g_t\|^2 \|d_t\|^2 \log(C/\delta)$ subgaussian and mean 0 given $\mathcal{F}_t$. Thus by the (subgaussian) Hoeffding inequality with probability at least $1 - \delta$ for any θ,

$$\left|\sum_{t=0}^{\tau-1} Q_t\right| \leq c\theta \sum_{t=0}^{\tau-1} \log(c/\delta) \|g_t\|^2 \|d_t\|^2 + \frac{1}{\theta} \log\Big(\frac{c}{\delta}\Big)$$

Now $\sum_{t=0}^{\tau-1} Q_t = E_\tau$ on $\bigcap A_t$. Notice, since $\mathbb{P}(A_t) \geq 1 - \delta$, $\mathbb{P}(\bigcap A_t) \geq 1 - \tau\delta$. Therefore with probability at least $1 - (\tau + 1)\delta$, E_τ satisfies the same bound as above. Rescaling δ to $\delta/(\tau + 1)$ and we have completed the proof. $\square$

This above lemma is the key for making the argument work. Because $V_x(y) \geq \frac{1}{2}\|x - y\|^2$, and recalling that f is ρ-Lipschitz continuous, we have

$$\left|\sum_{t=0}^{\tau-1} \alpha \langle (I - p_t p_t^\top) \nabla f(x_t), x_t - u \rangle\right| \leq \alpha \sqrt{c \sum_{t=0}^{\tau-1} \log^{3/2}(c\tau/\delta) \rho^2 V_{x_t}(u)}$$

after rescaling c and by choosing the optimal θ according to AM-GM $(x+y)/2 \geq \sqrt{xy}$. So, it remains to bound $\sum V_{x_t}(u)$. We can do this because Lemma 1.5 actually tells us a bound on $V_{x_\tau}(x^*)$.

So, by telescoping Equation (1.1), we arrive at the following lemma.

Lemma 1.10. *Define $\chi = \log\bigl(cT^2/\delta\bigr)$. Then with probability at least $1-\delta$, for every $1 \leq \tau \leq T$,*

$$V_{x_\tau}(x^*) \leq \frac{c\tau d\chi^{3/2}\alpha^2}{2}\rho^2 + V_{x_0}(x^*) + \alpha\rho\sqrt{c\sum_{t=0}^{\tau-1}\chi^{3/2}V_{x_t}(x^*)}$$

Proof. Telescoping Equation (1.1), we get

$$\begin{aligned}\sum_{t=0}^{\tau-1}\alpha(f(x_t)-f(x^*)) \leq{}& \sum_{t=0}^{\tau-1}\frac{\alpha^2}{2}\left\|\tilde{\nabla}f(x_t)\right\|^2 + V_{x_0}(x^*) - V_{x_\tau}(x^*)\\ &+\sum_{t=0}^{\tau-1}\alpha\langle(I-p_tp_t^\top)\nabla f(x_t), x_t - x^*\rangle\end{aligned}$$

Therefore, by the previous discussion, by Lemma 1.8 and using that $f(x_t) \geq f(x^*)$ for any x_t,

$$V_{x_\tau}(x^*) \leq \frac{c\tau d\chi^{3/2}\alpha^2}{2}\rho^2 + V_{x_0}(x^*) + \alpha\rho\sqrt{c\sum_{t=0}^{\tau-1}\log^{3/2}(c\tau/\delta)V_{x_t}(x^*)}$$

We can have this hold jointly for all $1 \leq \tau \leq T$ by rescaling $\delta \leftarrow \delta/T$. Then by using $\tau \leq T$, we arrive at the statement of the lemma. □

Naturally, this above lemma should tell us how much $V_{x_t}(x^*)$ can grow. This next lemma tells us an upper bound on how much that is.

Lemma 1.11. *Let Θ be any upper bound on $V_{x_0}(x^*)$. Then for some absolute constant c,*

$$\sup_{0\leq t\leq T} V_{x_t}(x^*) \leq 2\Theta + cd\alpha^2\rho^2\chi^{3/2}T$$

Proof. Define $M = \sup_{0\leq t\leq T} V_{x_t}(x^*)$. Notice that the right hand side is increasing in τ. Therefore applying sup to both sides, we get

$$\begin{aligned}M &\leq \frac{cTd\chi^{3/2}\alpha^2}{2}\rho^2 + \Theta + \alpha\rho\sqrt{c\sum_{t=0}^{T-1}\chi^{3/2}V_{x_t}(x^*)}\\ &\leq \frac{cTd\chi^{3/2}\alpha^2}{2}\rho^2 + \Theta + \alpha\rho\sqrt{cT\chi^{3/2}M}\end{aligned}$$

Using Young's inequality, we know that

$$\alpha\rho\sqrt{cT\chi^{3/2}M} \leq \frac{1}{2}M + c\alpha^2\rho^2\chi^{3/2}T$$

Therefore,

$$\begin{aligned} M &\leq cTd\chi^{3/2}\alpha^2\rho^2 + 2\Theta + 2c\alpha^2\rho^2\chi^{3/2}T \\ &\leq 2\Theta + cd\alpha^2\rho^2\chi^{3/2}T \end{aligned}$$

for some absolute constant c as desired. □

A natural question is: Is this a good bound at all? Shouldn't x_t be getting closer to the minmizer x^*? But we showed $V_{x_t}(x^*)$ grows linearly?

To answer this, recall from before that $\alpha = \sqrt{2\Theta/\rho^2 T}$. Ignoring the polylog factor $\chi^{3/2}$, and dimension d, we have actually shown that $V_{x_t}(x^*)$ stays bounded. But still, shouldn't it be decreasing to 0? Actually, it should not, since f could have many minimizers. The best bound you could hope to satisfy for every f is that it is just bounded.

For a similar choice of α, this tells us the accumulated error

$$\sum_{t=0}^{T-1} \alpha\langle (I - p_t p_t^\top)\nabla f(x_t), x_t - x^*\rangle$$

is around the same size as $V_{x_0}(x^*)$. This observation is what let's us prove convergence of Zero-Order mirror descent.

Theorem 1.12 (Zero-Order Mirror Descent Convergence)**.** *Let $\bar{x} = 1/T\sum_{t=0}^{T-1} x_t$, if we choose*

$$\alpha = \sqrt{\frac{2\Theta}{\rho^2 c\chi^{3/2}Td}}$$

then with probability at least $1-\delta$,

$$f(\bar{x}) - f(x^*) \leq \frac{1}{T\alpha}\cdot 9\Theta = \widetilde{O}\left(\rho\sqrt{\frac{\Theta d}{T}}\right)$$

Proof. By telescoping and using $V_{x_T}(x^*) \geq 0$, we get

$$\sum_{t=0}^{T-1} \alpha(f(x_t) - f^*) \leq \frac{cTd\chi^{3/2}\alpha^2}{2}\rho^2 + \Theta + \alpha\sqrt{c\sum_{t=0}^{T-1}\chi^{3/2}\rho^2 V_{x_t}(x^*)}$$

By using the previous lemma, we know that

$$\alpha\sqrt{c\sum_{t=0}^{T-1}\chi^{3/2}\rho^2 V_{x_t}(x^*)} \leq \alpha\sqrt{cT\chi^{3/2}\rho^2(2\Theta + c\alpha^2\rho^2\chi^{3/2}T)}$$

By choosing $\alpha = \sqrt{2\Theta/\rho^2 c\chi^{3/2}Td} \leq \sqrt{2\Theta/\rho^2 c\chi^{3/2}T}$ as $d \geq 1$, we know that

$$c\alpha^2\rho^2\chi^{3/2}T \leq 2\Theta$$

and therefore,

$$\alpha\sqrt{cT\chi^{3/2}\rho^2(2\Theta + c\alpha^2\rho^2\chi^{3/2}T)} \leq 4\Theta$$

Now again by our choice of α,

$$\frac{cTd\chi^{3/2}\alpha^2}{2}\rho^2 \leq 4\Theta$$

Therefore,

$$\alpha(f(\bar{x}) - f^*) \leq \frac{9\Theta}{T}$$

We conclude,

$$f(\bar{x}) - f^* \leq \frac{9\Theta}{T\alpha} = \rho c\sqrt{\frac{9}{2}}\sqrt{\frac{d\Theta}{T}}\chi^{3/4} = \widetilde{O}\left(\rho\sqrt{\frac{d\Theta}{T}}\right).$$

□

Chapter 2

Accelerated Zeroth-Order Gradient Descent via Linear Mixing

Assumption 2.1. We assume that the smoothness constant of f is $L \geq 1$. Otherwise, we can replace it with 1. Also assume that $0 < \delta < 1/e$.

We use the linear mixing framework to design an accelerated zero-order gradient descent algorithm. The idea is to query gradient descent y_t and mirror descent z_t at the same point x_t, and then linearly mix the answers $x_{t+1} = \lambda z_t + (1-\lambda) y_t$. The algorithm can be described as

$$
\begin{aligned}
x_{t+1} &= \lambda y_t + (1-\lambda) z_t \\
y_{t+1} &= x_{t+1} - \eta \tilde{\nabla} f(x_{t+1}) \\
z_{t+1} &= \mathrm{Mirr}_{z_t}(\alpha \tilde{\nabla} f(x_{t+1}))
\end{aligned}
$$

The first thing to do is to deal with quadratic error.

Lemma 2.2 (Quadratic Error)**.**

$$
\left\| \tilde{\nabla} f(x) - \frac{1}{r} \sum_{i=1}^{r} p_i p_i^\top \nabla f(x) \right\| \leq L\sigma \frac{1}{r} \sum_{i=1}^{r} \|p_i\|^3
$$

Proof. We have

$$
\begin{aligned}
\frac{f(x+\sigma p) - f(x-\sigma p)}{2\sigma} p - pp^\top \nabla f(x) &= \frac{1}{2\sigma} \left[\int_{-1}^{1} \frac{\mathrm{d}}{\mathrm{d}t} f(x + t\sigma p) - \sigma pp^\top \nabla f(x) dt \right] \\
&= \frac{1}{2\sigma} \left[\int_{-1}^{1} \langle \nabla f(x + t\sigma p) - \nabla f(x), \sigma p \rangle p dt \right]
\end{aligned}
$$

Now, by L-smoothness,

$$
\left| \int_{-1}^{1} \langle \nabla f(x + \sigma t p) - \nabla f(x), \sigma p \rangle p dt \right| \leq \int_{-1}^{1} L\sigma^2 |t| \, \|p\|^3 \, dt
$$

Therefore,

$$\left| \frac{f(x+\sigma p) - f(x-\sigma p)}{2\sigma} p - pp^\top \nabla f(x) \right| \leq L\sigma \|p\|^3$$

Then by triangle inequality,

$$\left\| \tilde{\nabla} f(x) - \frac{1}{r} \sum_{i=1}^{r} p_i p_i^\top \nabla f(x) \right\| \leq L\sigma \frac{1}{r} \sum_{i=1}^{r} \|p_i\|^3 .$$

□

Unfortunately, this bound is necessary. Ideally we would like to just say that $f(x+\sigma p) - f(x-\sigma p)/2\sigma p = pp^\top \nabla g(x)$ for some function $g(x)$ related to $f(x)$. Possibly a smoothed variant f_σ. But this cannot possible hold for every p unless f is quadratic in a neighborhood of x. To see this, we can use MVT to expand the LHS as $\langle \nabla f(x + \theta\sigma p), p\rangle p$. By taking $p = e_i$, we see that $e_i^\top \nabla g(x) = e_i^\top \nabla f(x + \theta_i \sigma e_i)$. Sending $\sigma \to 0$, $\nabla f(x+\theta_i\sigma e_i) \to \nabla f(x)$ under L-smoothness assumption, meaning that $\nabla g(x) = \nabla f(x)$. This would say that f is quadratic, which is not true in general.

2.1 Gradient Progress

Recall the following lemma from before.

Lemma 2.3. *Let X follow χ^2_d distribution and $0 < \delta < 1/e$. Then there exists L_δ such that for every d, with*

$$A := \left\{ dL_\delta^2 \leq X \leq 3d \log(C/\delta) \right\}$$

we have

$$\mathbb{E}[(X-d); A] = 0, \quad \mathbb{P}(A) \geq 1 - \delta$$

for some absolute constant C.

These lemmas tell us how much progress the gradient descent step is making.

Lemma 2.4 (Vershynin (2018) Corollary 7.3.2)**.** *Let $P \in \mathbb{R}^{d\times r}$ be a matrix with i.i.d. Gaussian entries. Define $\|A\|_{\mathrm{op}} = \sup_{u\in\mathbb{S}^{r-1}} \|Au\|_2$. Then*

$$\mathbb{P}\Big(\|P\|_{\mathrm{op}} \geq \sqrt{d} + \sqrt{r} + s \Big) \leq 2e^{-cs^2}$$

Proposition 2.5. *Define $\chi = \log(CrT/\delta)$ and let $P_t = \begin{bmatrix} p_{t,1} & \cdots & p_{t,r} \end{bmatrix}$. Define the bounded operator norm event*

$$\mathfrak{O}_t = \left\{ \frac{1}{\sqrt{r}} \|P\|_{\mathrm{op}} \leq C\frac{d}{r}\sqrt{\log(cTr/\delta)} \right\}$$

Then on $\mathfrak{O}_t$, for every $0 \le t \le T-1$,

$$\left\| \frac{1}{r} \sum_{i=1}^{r} p_{t,i} p_{t,i}^{\top} \nabla f(x_t) \right\|^2 \le \frac{Cd}{r} \log(cTr/\delta) \cdot \frac{1}{r} \sum_{i=1}^{r} |p_{t,i}^{\top} \nabla f(x_t)|^2$$

Furthermore,

$$\mathbb{P}\left(\bigcap_{t=0}^{T-1} \mathfrak{O}_t \right) \ge 1 - \delta$$

Proof. Write $P = \begin{bmatrix} p_{t,1} & \cdots & p_{t,r} \end{bmatrix} \in \mathbb{R}^{d \times r}$. By choosing $s = \sqrt{1/c \log(2/\delta)}$ above, we know that

$$\frac{1}{\sqrt{r}} \|P\|_{\mathrm{op}} \le 1 + \sqrt{\frac{d}{r}} + \frac{1}{c\sqrt{r}} \sqrt{\log(c/\delta)} \le C \sqrt{\frac{d}{r}} \sqrt{\log(c/\delta)}$$

with probability at least $1 - \delta$. As $PP^{\top} = \sum_{i=1}^{r} p_{t,i} p_{t,i}^{\top}$, we see that

$$\left\| \frac{1}{r} PP^{\top} \nabla f(x_t) \right\|^2 \le \left(\frac{1}{\sqrt{r}} \|P\|_{\mathrm{op}} \right)^2 \frac{1}{r} \left\| P^{\top} \nabla f(x_t) \right\|^2 \le \frac{Cd}{r} \log(c/\delta) \frac{1}{r} \sum_{i=1}^{r} |p_{t,i}^{\top} \nabla f(x_t)|^2$$

We then rescale δ to δ / Tr to complete the proof. □

Lemma 2.6. *Define the big progress event*

$$\mathfrak{P}_t = \left\{ \frac{1}{r} \sum_{i=1}^{r} |\langle p_{t,i}, \nabla f(x_t) \rangle|^2 \ge \frac{1}{2} \|\nabla f(x_t)\|^2 \right\}$$

As long as $r \ge 16\chi$, we have

$$\mathbb{P}\Big(\bigcap \mathfrak{P}_t \Big) \ge 1 - \delta.$$

Proof. As $\langle p_{t,i}, \nabla f(x_t) \rangle \overset{i.i.d.}{\sim} \mathcal{N}(0, \|\nabla f(x_t)\|^2)$, by the Laurent-Massart inequality (Laurent and Massart, 2000, Lemma 1, Equation (4.4)),

$$\mathbb{P}\Big(\chi_r^2 \le \frac{r}{2} \Big) \le \exp\Big(-\frac{r}{16} \Big)$$

Therefore as long as $r \ge 16\chi \ge 16 \log(T/\delta)$,

$$\frac{1}{r} \sum_{i=1}^{r} |\langle p_{t,i}, \nabla f(x_t) \rangle|^2 \ge \frac{1}{2} \|\nabla f(x_t)\|^2$$

for all $0 \le t \le T-1$. □

Discussion. It is unfortunate to necessitate a lower bound on r which will eventually grow with ε. The way around this is usually to instead look at the gradient progress over a time interval I of length $\asymp \log(T/\delta)$, in which case we could apply the same bound as above. However, this would require the gradients to be of roughly the same size, otherwise one gradient could dominate and we would only really have r effective samples even over the whole interval (compared to $r\log(T/\delta)$).

When proving gradient descent converges to a first-order stationary point (in the sequel), this is possible, and alleviates this requirement on r. However, in the mixing case, since the z_t point depends only on z_{t-1} and not on y_{t-1}, the mirrored sequence could drift extremely far from the gradient sequence. In this case, $x_{t+1} - x_t = (1-\lambda)(y_t - x_t) + \lambda z_t - \lambda x_t$ could potentially grow unbounded. For example, consider the one-variable function $f(x) = Gx + \frac{L}{2}x^2$. Assume that $x_t = 0$, we know that $\nabla f(x_t) = G$. Then $y_t = -\eta G$. $z_t = z_{t-1} - \alpha G$. Then

$$|x_{t+1} - x_t| = |x_{t+1}| = |(1-\lambda) - \eta G + \lambda(z_{t-1} - \alpha G)|$$

Sending $|z_{t-1}| \to \infty$, this gap can become unbounded. The α dictating z_{t-1}'s progression grows with ε, so this is (a priori) not an unrealistic scenario.

Lemma 2.7. *Define the bounded-perturbation event*

$$\mathfrak{B}_{t+1} \coloneqq \bigcap_{i=1}^{r} \left\{ \sqrt{dL_{\delta/rT}} \leq \|p_{t+1,i}\| \leq \sqrt{3d\chi} \right\}$$

Then on $\mathfrak{B}_{t+1}$, as long as $\eta \leq r/(12Ld\chi)$, we have

$$f(y_{t+1}) \leq f(x_{t+1}) - \frac{\eta}{8}\|\nabla f(x_{t+1})\|^2 + C'L^2\sigma^2\chi^3 d^3$$

Furthermore,

$$\mathbb{P}\left(\bigcap_{t=0}^{T-1} \mathfrak{B}_t\right) \geq 1 - \delta$$

Proof. By L-smoothness,

$$\begin{aligned} f(y_{t+1}) &\leq f(x_{t+1}) + \langle \nabla f(x_{t+1}), y_{t+1} - x_{t+1}\rangle + \frac{L}{2}\|y_{t+1} - x_{t+1}\|^2 \\ &= f(x_{t+1}) - \eta\langle \nabla f(x_{t+1}), \tilde{\nabla} f(x_{t+1})\rangle + \frac{L\eta^2}{2}\left\|\tilde{\nabla} f(x_{t+1})\right\|^2 \end{aligned}$$

Writing $\tilde{\nabla} f(x_{t+1}) = \frac{1}{r}\sum_{i=1}^{r} p_i p_i^\top \nabla f(x_{t+1}) + R_{t+1}$, we get

$$\begin{aligned} f(y_{t+1}) \leq f(x_{t+1}) &- \frac{\eta}{r}\sum_{i=1}^{r} |p_i^\top \nabla f(x_{t+1})|^2 - \eta\langle \nabla f(x_{t+1}), R_{t+1}\rangle \\ &+ L\eta^2 \left\| \frac{1}{r}\sum_{i=1}^{r} p_i p_i^\top \nabla f(x_{t+1})\right\|^2 + L\eta^2\|R_{t+1}\|^2 \end{aligned}$$

Now, $2\langle \eta\nabla f(x_{t+1}), R_{t+1}\rangle \leq \eta^2 \left\|\nabla f(x_{t+1})\right\|^2 + R_{t+1}^2$ by Cauchy-Schwarz and Young's inequality.

Define the event where p_i are bounded:

$$\mathfrak{B}_{t+1} \coloneqq \bigcap_{i=1}^{r} \left\{ \sqrt{dL_{\delta/rT}} \leq \|p_{t+1,i}\| \leq \sqrt{3d\chi} \right\}$$

We know that $\mathbb{P}(\mathfrak{B}_{t+1}) \geq 1 - \delta/rT$. Thus,

$$\begin{aligned} f(y_{t+1}) \leq f(x_{t+1}) &- \frac{\eta}{r}\sum_{i=1}^{r} |p_i^\top \nabla f(x_{t+1})|^2 + \eta^2\gamma \left\|\nabla f(x_{t+1})\right\|^2 + (L\eta^2+\gamma)\left\|R_{t+1}\right\|^2 \\ &+ \frac{CLd\eta^2\log(Tr/\delta)}{r}\cdot\frac{1}{r}\sum_{i=1}^{r}|p_i^\top \nabla f(x_{t+1})|^2 \end{aligned}$$

From quadratic error lemma we know that $|R_{t+1}| \leq 3L\sigma\log^{3/2}(1/\delta)d^{3/2}$. Now by choosing $\eta \leq r/(4CLd\chi)$, we know that $-\eta + CLd\chi\eta^2/r \leq -3/4\eta$. As $L\eta^2 < 1$ as $L \geq 1$, possibly changing the absolute constant C', we get,

$$f(y_{t+1}) \leq f(x_{t+1}) - \frac{3\eta}{4}\frac{1}{r}\sum_{i=1}^{r}|p_i^\top \nabla f(x_{t+1})|^2 + \eta^2\left\|\nabla f(x_{t+1})\right\|^2 + C'L^2\sigma^2\chi^3d^3$$

As long as $\eta \leq 1/8$, $-3\eta/8 + \eta^2 \leq -\eta/4$, so using the previous lemma, we ensure

$$f(y_{t+1}) \leq f(x_{t+1}) - \frac{\eta}{8}\left\|\nabla f(x_{t+1})\right\|^2 + C'L^2\sigma^2\chi^3d^3$$

$\square$

2.2 Mirror Descent against Progress

We can copy the mirror descent lemma from last time:

Lemma 2.8. *If $z_{t+1} = \mathrm{Mirr}_{z_t}(\alpha\tilde{\nabla} f(x_{t+1}))$, then*

$$\forall u \in \mathbb{R}^d \quad \alpha\langle\tilde{\nabla} f(x_{t+1}), z_t - u\rangle \leq \frac{\alpha^2}{2}\left\|\tilde{\nabla} f(x_{t+1})\right\|^2 + V_{z_t}(u) - V_{z_{t+1}}(u).$$

The standard coupling lemma comes from using $\left\|\nabla f(x_{t+1})\right\|^2 \leq 2L(f(x_{t+1}) - f(y_{t+1}))$. However this doesn't work here, as we are using an estimator. But we can fix that using the following lemma.

Lemma 2.9. *Define the event*

$$\mathfrak{G}_{t+1} = \left\{ \left\| \frac{1}{r}\sum_{i=1}^{r} p_{t+1,i}p_{t+1,i}^\top \nabla f(x_{t+1}) \right\|^2 \leq \frac{Cd}{r}\chi\left\|\nabla f(x_{t+1})\right\|^2 \right\}$$

On $\mathfrak{B}_{t+1} \cap \mathfrak{G}_{t+1} \cap \mathfrak{O}_{t+1} \cap \mathfrak{P}_{t+1}$, we have for all slack $\gamma \geq 2$,

$$\frac{Cd\gamma\chi}{r} \|\nabla f(x_{t+1})\|^2 + \left\|\tilde{\nabla} f(x_{t+1})\right\|^2 \leq \frac{16Cd\gamma\chi}{r\eta} \left(f(x_{t+1}) - f(y_{t+1}) + C'L^2\sigma^2\chi^3 d^3\right)$$

Furthermore,

$$\mathbb{P}\Big(\bigcap \mathfrak{G}_t\Big) \geq 1 - \delta.$$

Proof. Define the event

$$\mathfrak{G}_{t+1} = \left\{\left\|\frac{1}{r}\sum_{i=1}^{r} p_{t+1,i} p_{t+1,i}^\top \nabla f(x_{t+1})\right\|^2 \leq \frac{Cd}{r}\chi \|\nabla f(x_{t+1})\|^2\right\}$$

Then on $\mathfrak{G}_{t+1}$,

$$\begin{aligned}
\left\|\tilde{\nabla} f(x_{t+1})\right\|^2 &\leq 2\left\|\frac{1}{r}\sum_{i=1}^{r} p_i p_i^\top \nabla f(x_{t+1})\right\|^2 + 2\|R_{t+1}\|^2 \\
&\leq 2\frac{Cd}{r}\chi \cdot \|\nabla f(x_{t+1})\|^2 + 2\|R_{t+1}\|^2 \\
&\leq \frac{2Cd}{r}\chi \|\nabla f(x_{t+1})\|^2 + C'L^2\sigma^2 \log^3(rT/\delta) d^3
\end{aligned}$$

We will need some slack of the gradient norm to tune later. For $\gamma \geq 2$, as $\gamma - \gamma/2 \geq 1$, this is at most

$$\frac{2Cd\gamma}{r}\chi \|\nabla f(x_{t+1})\|^2 + C'L^2\sigma^2 \log^3(rT/\delta) d^3 - \frac{Cd\gamma}{r}\chi \|\nabla f(x_{t+1})\|^2$$

From before we know that

$$\begin{aligned}
&\frac{\eta}{8} \|\nabla f(x_{t+1})\|^2 \\
&\leq f(x_{t+1}) - f(y_{t+1}) + C'L^2\sigma^2 \log^3(rT/\delta) d^3
\end{aligned}$$

Therefore an upper bound on $\left\|\tilde{\nabla} f(x_{t+1})\right\|^2$ is

$$\begin{aligned}
&\frac{16Cd\gamma\chi}{r\eta}\left[f(x_{t+1}) - f(y_{t+1}) + C'L^2\sigma^2 \log^3(rT/\delta) d^3\right] \\
&+ C'L^2\sigma^2\chi^3 d^3 - \frac{Cd\gamma}{r}\chi \|\nabla f(x_{t+1})\|^2
\end{aligned}$$

As $\eta < 1/8$ we can absorb the last term into the first by changing the constant C'. Finally, after an application of the triangle inequality and Lemma 1.9, we have $\mathbb{P}(\mathfrak{G}_{t+1}) \geq 1 - \delta/T$, which gives the statement of the lemma. $\square$

2.3 Mirror Descent and Linear Mixing

This lemma is the same as before.

Lemma 2.10. *When λ is chosen so $\frac{1-\lambda}{\lambda} = \frac{8Cd\gamma\chi\alpha}{\eta}$, the "regret" is bounded by*

$$\alpha\langle \nabla f(x_{t+1}), x_{t+1} - u\rangle - \alpha\langle \nabla f(x_{t+1}), z_t - u\rangle \leq \frac{8Cd\gamma\chi\alpha^2}{\eta}(f(y_t) - f(x_{t+1}))$$

Proof. We have by convexity and the definition of $x_{t+1} = \lambda z_t + (1-\lambda)y_t$,

$$\begin{aligned}
&\alpha\langle \nabla f(x_{t+1}), x_{t+1} - u\rangle - \alpha\langle \nabla f(x_{t+1}), z_t - u\rangle \\
&= \alpha\langle \nabla f(x_{t+1}), x_{t+1} - z_t\rangle = \frac{(1-\lambda)\alpha}{\lambda}\langle \nabla f(x_{t+1}), y_t - x_{t+1}\rangle \\
&\leq \frac{(1-\lambda)\alpha}{\lambda}(f(y_t) - f(x_{t+1}))
\end{aligned}$$

Now let $\lambda \in (0,1)$ satisfy $\frac{1-\lambda}{\lambda} = \frac{8Cd\gamma\chi\alpha}{\eta}$ and we are done. □

This lemma reveals the martingale term we need to bound.

Lemma 2.11.

$$\begin{aligned}
\alpha\langle \nabla f(x_{t+1}), z_t - u\rangle \leq \alpha\langle \tilde{\nabla} f(x_{t+1}), z_t - u\rangle + \alpha\sum_{i=1}^{r}\langle (I - p_i p_i^\top)\nabla f(x_{t+1}), z_t - u\rangle \\
+ C'T\alpha^2 \left\|R_{t+1}\right\|^2 + \frac{1}{4T}V_{z_t}(u).
\end{aligned}$$

Proof.

$$\langle \nabla f(x_{t+1}), z_t - u\rangle = \langle \tilde{\nabla} f(x_{t+1}), z_t - u\rangle + \sum_{i=1}^{r}\langle (I - p_i p_i^\top)\nabla f(x_{t+1}), z_t - u\rangle + \langle R_{t+1}, z_t - u\rangle$$

Now, by Young's inequality, $\alpha\langle R_{t+1}, z_t - u\rangle \leq C'\alpha^2 T\left\|R_{t+1}\right\|^2 + \frac{1}{8T}\left\|z_t - u\right\|^2$, by strong convexity of Bregman divergence, $V_{z_t}(u) \geq \frac{1}{2}\left\|z_t - u\right\|^2$, so we arrive at the statement of the lemma. □

2.4 Dealing with the Martingale

The main idea from here is that, the martingale term

$$\sum_{t=0}^{T-1}\sum_{i=1}^{r}\langle (I - p_i p_i^\top)\nabla f(x_{t+1}), z_t - u\rangle$$

can be bounded by the "Energy" $V_{z_t}(x^*)$, which itself can be bounded by the sum of the gradient norms $\sum\left\|\nabla f(x_t)\right\|^2$ and $\Theta \geq V_{z_0}(x^*)$.

This highlights a very general principle. To deal with martingale error terms with high probability, try bounding them by some other part of the equation which should have typical size $\Theta(1)$. In this case, that will be $V_{z_t}(x^*)$ for us.

First, we highlight some intuition. Due to oscillatory cancellation, the typical size of the martingale term is much smaller than any deterministic inequality would suggest. This is because it is a martingale difference sequence.

Definition 2.12 (Martingale Difference Sequence)**.** $\{\xi_t\}_{t\geq 1}$ is a martingale difference sequence wrt filtration $\mathcal{F}_t$ if $\xi_t \in \mathcal{F}_t$ and

$$\mathbb{E}[\xi_t \mid \mathcal{F}_{t-1}] = 0$$

The main intuition is that in a martingale difference sequence, the variance of the full sum is the sum of the variance of each ξ_t, which is vastly superior than including all the cross covariance terms $\mathrm{Cov}(\xi_i, \xi_j)$. This improves our bound substantially.

Lemma 2.13. *Define* $S_T = \sum_{t=1}^T \xi_t$*. Then*

$$\mathbb{E}\left[S_T^2\right] = \sum_{t=1}^T \mathbb{E}\left[\xi_t^2\right]$$

Proof. We have

$$\mathbb{E}\left[M_T^2\right] = \sum \mathbb{E}\left[\xi_t^2\right] + 2\mathbb{E}\left[\sum_{i<j} \xi_i \xi_j\right]$$

Now,

$$\mathbb{E}\left[\sum_{i<j} \xi_i \xi_j\right] = \sum_{i<j} \mathbb{E}[\mathbb{E}[\xi_i \xi_j \mid \mathcal{F}_i]] = \sum_{i<j} \mathbb{E}[\xi_i \mathbb{E}[\xi_j \mid \mathcal{F}_i]] = 0.$$

□

Lemma 2.14. *Define*

$$\mathfrak{M}_{t+1} = \left\{ \left| \sum_{t=0}^{\tau-1} \sum_{i=1}^{r} \alpha \langle (I - p_i p_i^\top) \nabla f(x_{t+1}), z_t - u \rangle \right| \leq \alpha \sqrt{Cr \sum_{t=0}^{\tau-1} \chi \left\| \nabla f(x_t) \right\|^2 V_{z_t}(u)} \right\}$$

Then

$$\mathbb{P}\left(\bigcap_{t=0}^{T-1} \mathfrak{M}_t \right) \geq 1 - \delta.$$

Proof. From Lemma 1.9 above, we know that

$$\begin{aligned}
&\left|\sum_{t=0}^{\tau-1}\sum_{i=1}^{r}\alpha\langle(I-p_ip_i^\top)\nabla f(x_{t+1}),z_t-u\rangle\right| \\
&\leq C\theta\sum_{t=0}^{\tau-1}\sum_{i=1}^{r}\log(CrT/\delta)\left\|\nabla f(x_{t+1})\right\|^2\left\|z_t-u\right\|^2+\frac{1}{\theta}\log\left(\frac{CrT}{\delta}\right)
\end{aligned}$$

Using $V_{z_t}(u)\geq\frac{1}{2}\left\|z_t-u\right\|^2$ and balancing the two terms by carefully choosing θ we arrive at the statement of the lemma. □

2.5 Putting it all Together

First, as in the proof of Mirror Descent, we bound the regret, then $V_{z_t}(x^*)$, then last we will tune α.

Lemma 2.15 (Regret Bound)**.** *On $\bigcap_{t=1}^{T}(\mathfrak{B}_t\cap\mathfrak{O}_t\cap\mathfrak{P}_t\cap\mathfrak{G}_t\cap\mathfrak{M}_t)$, we have*

$$\begin{aligned}
&\sum_{t=0}^{\tau-1}\alpha\langle\nabla f(x_t),x_t-x^*\rangle \\
&\leq\frac{8Cd\gamma\chi}{r\eta}\alpha^2(f(y_0)-f(y_\tau))-\frac{C\alpha^2d\gamma\chi}{2r}\sum_{t=0}^{\tau-1}\left\|\nabla f(x_{t+1})\right\|^2 \\
&+\alpha\sqrt{\frac{C}{r}\sum_{t=0}^{\tau-1}\chi\left\|\nabla f(x_{t+1})\right\|^2V_{z_t}(x^*)}+\frac{C'(\alpha^2T+\alpha T^2)\gamma L^2\sigma^2\chi^4d^4}{r\eta} \\
&+V_{z_0}(x^*)-V_{z_\tau}(x^*)+\frac{1}{4T}\sum_{t=0}^{\tau-1}V_{z_t}(x^*)
\end{aligned}$$

Proof. By Lemma 2.10, we know that

$$\sum_{t=0}^{\tau-1}\alpha\langle\nabla f(x_t),x_t-x^*\rangle\leq\sum_{t=0}^{\tau-1}\frac{8Cd\gamma\chi\alpha^2}{\eta}(f(y_t)-f(x_{t+1}))+\sum_{t=0}^{\tau-1}\alpha\langle\nabla f(x_{t+1}),z_t-u\rangle$$

By Lemma 2.11, we know that

$$\begin{aligned}
&\sum_{t=0}^{\tau-1}\alpha\langle\nabla f(x_{t+1}),z_t-u\rangle \\
&\leq\alpha\sum_{t=0}^{\tau-1}\left[\langle\tilde{\nabla}f(x_{t+1}),z_t-u\rangle+\frac{1}{r}\sum_{i=1}^{r}\langle(I-p_ip_i^\top)\nabla f(x_{t+1}),z_t-u\rangle+2T\left\|R_{t+1}\right\|^2+\frac{1}{T}V_{z_t}(u)\right]
\end{aligned}$$

From above we know that

$$\alpha\sum_{t=0}^{\tau-1}\sum_{i=1}^{r}\langle(I-p_ip_i^\top)\nabla f(x_{t+1}),z_t-u\rangle\leq\alpha\sqrt{Cr\sum_{t=0}^{\tau-1}\chi\left\|\nabla f(x_{t+1})\right\|^2V_{z_t}(u)}$$

From Lemmas 2.8 and 2.9 we know that

$$\begin{aligned}
&\sum_{t=0}^{\tau-1} \alpha\langle \tilde{\nabla} f(x_{t+1}), z_t - u\rangle \\
&\leq \alpha^2 \sum_{t=0}^{\tau-1} \frac{8Cd\gamma\chi}{r\eta}\left(f(x_{t+1}) - f(y_{t+1}) + C'L^2\sigma^2\chi^3 d^3\right) - \frac{C\alpha^2 d\gamma\chi}{2r}\sum_{t=0}^{\tau-1}\|\nabla f(x_{t+1})\|^2 \\
&+ V_{z_0}(x^*) - V_{z_\tau}(x^*)
\end{aligned}$$

Recall that $\|R_{t+1}\|^2 \leq C'L^2\sigma^2\chi^3 d^3$. Putting all of it together, we get:

$$\begin{aligned}
&\sum_{t=0}^{\tau-1} \alpha\langle \nabla f(x_t), x_t - x^*\rangle \\
&\leq \frac{8Cd\gamma\chi}{r\eta}\alpha^2(f(y_0) - f(y_\tau)) - \frac{C\alpha^2 d\gamma\chi}{2r}\sum_{t=0}^{\tau-1}\|\nabla f(x_{t+1})\|^2 \\
&+ \alpha\sqrt{\frac{C}{r}\sum_{t=0}^{\tau-1}\chi\|\nabla f(x_{t+1})\|^2 V_{z_t}(x^*)} + \frac{C'\alpha^2 T\gamma L^2\sigma^2\chi^4 d^4}{r\eta} + C'\alpha T^2 L^2\sigma^2\chi^3 d^3 \\
&+ V_{z_0}(x^*) - V_{z_\tau}(x^*) + \frac{1}{4T}\sum_{t=0}^{\tau-1} V_{z_t}(x^*)
\end{aligned}$$

By using $r \leq d$ and $\eta < 1$ to handle the two quadratic terms we arrive at the statement of the lemma. □

This next lemma is what we meant by bounding the martingale by other already-present $\Theta(1)$ terms.

Lemma 2.16 (Energy Bound). *On $\bigcap_{t=1}^{T}(\mathfrak{B}_t \cap \mathfrak{D}_t \cap \mathfrak{P}_t \cap \mathfrak{G}_t \cap \mathfrak{M}_t)$, upon defining $M = \sup_{0\leq t\leq T-1} V_{z_t}(x^*)$, choosing the slack $\gamma = 8$ and letting $\Theta \geq V_{z_0}(x^*)$ be any upper bound, we have*

$$\frac{M}{2} \leq \frac{256Cd\chi}{r\eta}\alpha^2(f(y_0) - f^*) + \Theta + \frac{C'(\alpha^2 T + \alpha T^2)L^2\sigma^2\chi^4 d^4}{r\eta}$$

Proof. By convexity we know that $f^* = f(x^*) \geq f(x) + \langle \nabla f(x), x^* - x\rangle$. Rearranging, we get $\langle \nabla f(x), x^* - x\rangle \geq f(x) - f^* \geq 0$. Using Young's inequality, we know that

$$\alpha\sqrt{\frac{C}{r}\chi\sum_{t=0}^{\tau-1}\|\nabla f(x_{t+1})\|^2 V_{z_t}(x^*)} \leq \frac{M}{4} + \alpha^2\frac{4C\chi}{r}\sum_{t=0}^{\tau-1}\|\nabla f(x_{t+1})\|^2$$

Therefore by the previous lemma, and using $\frac{1}{T}\sum_{t=0}^{\tau-1} V_{z_\tau}(x^*) \leq \frac{1}{4}M$, we know that

$$\begin{aligned}V_{z_\tau}(x^*) \leq \frac{8C\gamma d\chi}{r\eta}\alpha^2(f(y_0)-f(y_\tau)) - \frac{C\alpha^2 d\gamma\chi}{2r}\sum_{t=0}^{\tau-1}\|\nabla f(x_{t+1})\|^2 \\ + V_{z_0}(x^*) + \frac{M}{4} + \frac{C'(\alpha^2 T+\alpha T^2)\gamma L^2\sigma^2\chi^4 d^4}{r\eta} \\ + \alpha^2\frac{4C\chi}{r}\sum_{t=0}^{\tau-1}\|\nabla f(x_{t+1})\|^2 + \frac{M}{4}\end{aligned}$$

By choosing $\gamma = 8$, as $d \geq 1$, we know that

$$-\frac{C\alpha^2 d\gamma\chi}{2r}\sum_{t=0}^{\tau-1}\|\nabla f(x_{t+1})\|^2 + \alpha^2\frac{4C\chi}{r}\sum_{t=0}^{\tau-1}\|\nabla f(x_{t+1})\|^2 \leq 0$$

By using $f(y_0) - f(y_\tau) \leq f(y_0) - f^*$ along with $V_{z_0}(x^*) \leq \Theta$, and taking supremum of both sides, we have

$$M \leq \frac{64Cd\chi}{r\eta}\alpha^2(f(y_0)-f^*) + \Theta + \frac{M}{2} + \frac{C'(\alpha^2 T+\alpha T^2)L^2\sigma^2\chi^4 d^4}{r\eta}$$

which completes the proof. □

Theorem 2.17 (Halving Function Objective). *Assuming the initial error $f(x_0)-f^* \leq \Delta$, and defining $\overline{x} = 1/T\sum_{t=0}^{T-1} x_t$, by choosing*

$$\eta = \frac{r}{4CLd\chi}, \qquad \alpha = \frac{r}{16Cd\chi}\sqrt{\frac{\Theta}{L\Delta}}, \qquad \sigma^2 \leq \frac{r^3\Delta^{3/2}}{4C'L^{7/2}\chi^6 d^6\sqrt{\Theta}}, \qquad r \geq 16\chi$$

In

$$T = \frac{256Cd\chi}{r}\sqrt{\frac{L\Theta}{\Delta}}$$

iterations we have $f(\overline{x}) - f^* \leq \Delta/2$ with probability at least $1-5\delta$.

Proof. On $\bigcap_{t=1}^{T}(\mathfrak{B}_t \cap \mathfrak{O}_t \cap \mathfrak{P}_t \cap \mathfrak{G}_t \cap \mathfrak{M}_t)$ all our previous lemmas hold. Each has probability at least $1-\delta$ so the intersection of 5 has probability at least $1-5\delta$.

By Lemma 2.15, we know that

$$\begin{aligned}&\sum_{t=0}^{T-1}\alpha\langle\nabla f(x_t), x_t - x^*\rangle \\ &\leq \frac{64Cd\chi}{r\eta}\alpha^2(f(y_0)-f(y_T)) - \frac{C\alpha^2 d\gamma\chi}{2r}\sum_{t=0}^{T-1}\|\nabla f(x_{t+1})\|^2 \\ &+ \alpha\sqrt{\frac{C}{r}\sum_{t=0}^{T-1}\chi\|\nabla f(x_{t+1})\|^2 V_{z_t}(x^*)} + \frac{C'(\alpha^2 T+\alpha T^2)L^2\sigma^2\chi^4 d^4}{r\eta} \\ &+ V_{z_0}(x^*) - V_{z_T}(x^*) + \frac{1}{4T}\sum_{t=0}^{T-1} V_{z_t}(x^*)\end{aligned}$$

Using Young's inequality in exactly the way as the previous lemma, we can rearrange this to

$$\begin{aligned}&\sum_{t=0}^{T-1} \alpha\langle \nabla f(x_t), x_t - x^*\rangle \\ &\leq \frac{64Cd\chi}{r\eta}\alpha^2(f(y_0) - f^*) + \frac{C'(\alpha^2 T + \alpha T^2)L^2\sigma^2\chi^4 d^4}{r\eta} + \Theta + M/4\end{aligned}$$

Using the previous lemma, we have

$$\begin{aligned}&\sum_{t=0}^{T-1} \alpha\langle \nabla f(x_t), x_t - x^*\rangle \\ &\leq \frac{128Cd\chi}{r\eta}\alpha^2(f(y_0) - f^*) + \frac{C'(\alpha^2 T + \alpha T^2)L^2\sigma^2\chi^4 d^4}{r\eta} + 2\Theta\end{aligned}$$

Let $\overline{x} = \sum_{t=0}^{T-1} x_t$. By convexity,

$$T(f(\overline{x}) - f^*) \leq \sum_{t=0}^{T-1}(f(x_t) - f^*) \leq \sum_{t=0}^{T-1}\langle \nabla f(x_t), x_t - x^*\rangle$$

Therefore, assuming the initial objective gap $f(y_0) - f^* \leq \Delta$,

$$f(\overline{x}) - f^* \leq \frac{128Cd\chi\alpha\Delta}{r\eta T} + \frac{2\Theta}{\alpha T} + \frac{C'(\alpha + T)L^2\sigma^2\chi^4 d^4}{r\eta}$$

Now choose the learning rate $\eta = r/(4CLd\chi)$. Then we get

$$f(\overline{x}) - f^* \leq \frac{512C^2d^2L\chi^2\alpha\Delta}{r^2 T} + \frac{2\Theta}{\alpha T} + \frac{C'(\alpha + T)L^3\sigma^2\chi^5 d^5}{r^2}$$

By choosing

$$\alpha = \frac{r}{16Cd\chi}\sqrt{\frac{\Theta}{L\Delta}},$$

and

$$T = \frac{256Cd\chi}{r}\sqrt{\frac{L\Theta}{\Delta}}$$

as $L \geq 1$ we have $T \geq \alpha$, so the above is at most

$$\frac{\Delta}{4} + \sqrt{\frac{L\Theta}{\Delta}}\frac{C'L^3\sigma^2\chi^6 d^6}{r^3}$$

Then as long as

$$\sigma^2 \leq \frac{r^3 \Delta^{3/2}}{4C' L^{7/2} \chi^6 d^6 \sqrt{\Theta}}$$

We have

$$f(\overline{x}) - f^* \leq \frac{\Delta}{2}$$

thereby halving the objective value. This completes the proof. □

TODO: RESTARTS

Chapter 3

Nonconvex Zeroth-Order Optimization with EGGROLL

3.1 Problem Setup

Sarkar et al. (2025) introduce EGGROLL as a low-rank evolutionary strategies method for scalable black-box optimization of large neural networks, with major applications being LLM fine tuning and post-training. In this excerpt, we prove that (symmetric) EGGROLL converges to a first order stationary point.

Assumption 3.1 (Properties of f)**.** We suppose that $f : \mathbb{R}^{d\times m} \to \mathbb{R}$ satisfies the following properties:

1. f is differentiable and lower bounded, i.e. $f^* = \min_x f(x) > -\infty$
2. f is L-gradient Lipschitz, i.e.

$$\|\nabla f(X) - \nabla f(Y)\|_F \leq L \|X - Y\|_F \quad \forall X, Y \in \mathbb{R}^{d\times m}$$

Definition 3.2. A point $X \in \mathbb{R}^{d\times m}$ is an ε-first order stationary point if

$$\|\nabla f(X)\|_F \leq \varepsilon$$

In this excerpt we work with symmetric EGGROLL, following the low-rank perturbation framework of Sarkar et al. (2025).

Definition 3.3 (Symmetric EGGROLL Update)**.** We define a rank r symmetric zeroth order estimator as follows:

$$g_\sigma^r(X) \coloneqq \frac{1}{r}\sum_{i=1}^{r} \frac{f(X+\sigma P_i) - f(X-\sigma P_i)}{2\sigma} P_i$$

using $P_i = a_i b_i^\top$ where $a_i \overset{i.i.d.}{\sim} \mathcal{N}(0, I_d)$ and $b_i \overset{i.i.d.}{\sim} \mathcal{N}(0, I_m)$, and $\sigma > 0$ is the smoothing radius.

Algorithm 1 Rank-r Symmetric EGGROLL

Require: Initial point $X_0 \in \mathbb{R}^{d\times m}$, horizon T, stepsize $\eta > 0$, smoothing radius $\sigma > 0$, rank $r \in \mathbb{N}$
  for $t = 0, \ldots, T-1$ **do**
    Sample $\{a_{t,i}\}_{i=1}^r \sim \mathcal{N}(0, I)$ and $\{b_{t,i}\}_{i=1}^r \sim \mathcal{N}(0, I)$ to compute $g_\sigma^r(X_t)$
    Update $X_{t+1} \leftarrow X_t - \eta g_\sigma^r(X_t)$
  end for
  return X_T

We state an informal version of the main result.

Theorem 3.4. *Consider running Algorithm 1. Suppose* $0 < \delta \le 1/e$. *Suppose*

$$\sigma = \tilde{O}\Big(\frac{\varepsilon}{dm}\Big), \quad \eta = \tilde{O}\Big(\frac{r}{dm}\Big).$$

Then in

$$T = \tilde{\Omega}\left(\frac{dmL(f(X_0) - f^*)}{r\varepsilon^2}\right)$$

iterations (with each iteration using $2r$ *function evaluations), with probability at least* $1 - \delta$*, at least* $3/4$ *of the iterates are* ε*-first order stationary points.*

3.2 Proof Outline

First, we prove a few concentration bounds which will aid us in bounding the error terms. The lion's share of work went into proving Theorem 3.7, which is the main tool used to get the running time above. We then introduce a preliminary lemma on function decrease, Lemma 3.12. Because the update could either decrease function value if it aligns sufficiently with the gradient, or it could increase it but by an amount of the same order as other error terms, we use Lemma 3.16 and Lemma 3.18 to handle these two cases. Armed with these lemmata, Lemma 3.19 is the remaining work to be done, and we arrive at the formal verison of our main theorem, Theorem 3.20 after a careful choice of parameters.

3.3 Concentration Bounds

Here we prove a novel concentration bound about the EGGROLL error. Throughout this excerpt, all norms of matrices is the Frobenius norm

$$\|A\|_F = \left(\sum_{i,j} A_{i,j}^2\right)^{1/2}.$$

3.3.1 Novel Concentration Bound

Throughout this paper, we denote

$$\mathbb{E}_{\mathcal{F}_{t-1}}[X; A] := \mathbb{E}[X\mathbb{I}_A \mid \mathcal{F}_{t-1}]$$

We will need the following lemma from Jin et al. (2019).

Lemma 3.5. *Let random vectors $x_1, \dots, x_n \in \mathbb{R}^d$, and corresponding filtrations $\mathcal{F}_t = \sigma(X_1, \dots, X_t)$ satisfy that*

$$\mathbb{E}_{\mathcal{F}_{t-1}}[x_t] = 0, \quad \mathbb{P}_{\mathcal{F}_{t-1}}(\|x_t\| \geq s) \leq 2e^{-\frac{s^2}{2\sigma^2}} \quad \forall s \in \mathbb{R}$$

Then there exists an absolute constant c such for any fixed $\delta > 0, \theta > 0$, with probability at least $1 - \delta$,

$$\left\| \sum_{i=1}^{n} X_i \right\| \leq c \cdot \theta \sum_{i=1}^{n} \sigma_i^2 + \frac{1}{\theta} \log\left(\frac{2d}{\delta}\right).$$

This will help us later.

Lemma 3.6. *Let X follow χ_d^2 distribution and $0 < \delta < 1/e$. Then there exists L_δ such that for every d, with*

$$A := \left\{dL_\delta^2 \leq X \leq 3d\log(C/\delta)\right\}$$

we have

$$\mathbb{E}[(X - d); A] = 0, \quad \mathbb{P}(A) \geq 1 - \delta$$

for some absolute constant C.

Proof. Let $f_d(x)$ be the χ_d^2 density,

$$f_d(x) = \frac{1}{2^{d/2}\Gamma(d/2)} x^{d/2-1} e^{-x/2}, \quad x \geq 0.$$

A simple calculation shows that

$$(x - d)f_d(x) = -2\frac{\mathrm{d}}{\mathrm{d}x}(xf_d(x))$$

Define $U = 3d\log(1/\delta)$ and define the set

$$A = \{L \leq X \leq U\}$$

We want

$$\mathbb{E}[(X - d); A] = \int_L^U (x - d)f_d(x)dx = -2Lf_d(L) - 2Uf_d(U) \stackrel{\text{want}}{=} 0.$$

Therefore, we need

$$\begin{aligned} & L f_d(L) = U f_d(U) \\ \iff & L^d e^{-L} = U^d e^{-U} \\ \iff & U - L = d \log(U/L) \end{aligned}$$

Now define $\ell = L/d$ and $u = U/d$. Then we want to solve

$$\begin{aligned} & du - d\ell = d \log(u/\ell) \\ \iff & \ell e^{-\ell} = u e^{-u} \end{aligned}$$

The function $g(x) = xe^{-x}$ is increasing on $[0, 1)$ and decreasing on $(1, \infty)$. Therefore, as long as $u > 1$, which follows as $\delta < 1/e$, there exists a unique $\ell < 1$ satisfying the above equation. Now notice, as $e^{-\ell} \geq e^{-1}$,

$$\ell e^{-1} \leq \ell e^{-\ell} = 3 \log(1/\delta)\delta^3 \leq 3e^{-1}\delta^2$$

Now we have that

$$\mathbb{P}(A^c) = \mathbb{P}(X \geq U) + \mathbb{P}(X \leq L)$$

Recall the Chernoff bound For $0 < \alpha < 1$,

$$\mathbb{P}(X \leq \alpha d) \leq (\alpha e^{1-\alpha})^{d/2} \leq (e\alpha)^{d/2}$$

Then we have

$$\mathbb{P}(X \leq L) \leq \mathbb{P}\big(X \leq 3d\delta^2\big) \leq (3e\delta^2)^{d/2} = (\sqrt{3e}\delta)^d$$

When $\sqrt{3e}\delta < 1$, this is $< \sqrt{3e}\delta$, otherwise ths bound holds vacuously. Therefore $\mathbb{P}(X \leq L) \leq \sqrt{3e}\delta$. On the other hand, using that for $t < 1/2$,

$$\mathbb{E}\left[e^{t\|Z\|^2}\right] = (1 - 2t)^{-d/2},$$

along with Markov's inequality, we get

$$\mathbb{P}(X \geq 3d \log(1/\delta)) = \mathbb{P}(\exp(X/3) \geq \exp(d \log(1/\delta))) \leq 3^{d/2}\delta^d \leq \sqrt{3}\delta$$

And thus $\mathbb{P}(A) \geq 1 - C\delta$ for the absolute constant $C = \sqrt{3} + \sqrt{3e}$. □

Here is the novel concentration bound. The proof is inspired by Proposition 2 of Ren et al. (2023) but with many important differences.

Theorem 3.7. *Let $\{\mathcal{F}_t\}_{t\geq -1}$ be a filtration, and let $\{a_t\}_{t\geq 0}, \{b_t\}_{t\geq 0}$ be two sequences of independent random vectors such that $a_t \sim \mathcal{N}(0, I_d)$, $b_t \sim \mathcal{N}(0, I_m)$, $a_t, b_t \in \mathcal{F}_t$ and*

a_t, b_t are independent of $\mathcal{F}_{t-1}$. Let $\{V_t\}_{t\geq 0}$ be a sequence of random matrices such that $V_t \in \mathcal{F}_{t-1}$. For each $\tau \geq 0$, let

$$E_\tau = \sum_{t=0}^{\tau-1} \left(V_t - \langle V_t, a_t b_t^\top \rangle a_t b_t^\top\right)$$

Then, there exists some absolute constants $C > 0, c > 0$ such that for any $\tau \in \mathbb{Z}^+$, and $0 < \delta < 1/e$, for any $\theta > 0$, with probability at least $1 - \delta$ given $\mathcal{F}_{t-1}$,

$$\|E_\tau\|_F \leq c \cdot \theta \sum_{t=0}^{\tau-1} dm \log^3\left(\frac{C\tau m}{\delta}\right) \|V_t\|_F^2 + \frac{1}{\theta} \log\left(\frac{Cm^2 d\tau}{\delta}\right).$$

Proof. Let $V_t = P\Sigma Q^\top = \sum_{i=1}^r \sigma_i p_i q_i^\top$ be the singular value decomposition of V_t. Extend $\{q_1, \ldots, q_r\}$ to an orthonormal basis $\{q_1, \ldots, q_m\}$ of $\mathbb{R}^m$. Define the events

$$A_t = \left\{\sqrt{d} L_\delta < \|a_t\| < \sqrt{3d \log(C/\delta)}\right\}$$

and

$$B_{t,i} = \left\{L_\delta < |b_t^\top q_i| < \sqrt{3 \log(C/\delta)}\right\}, \quad B_t = \bigcap_{i=1}^m B_{t,i}.$$

Let R be an orthogonal matrix $RR^\top = R^\top R = I$, and notice that Ra has the same distribution as a, and that A_t is the same event when $\|a_t\|$ is replaced with $\|Ra_t\|$. Therefore,

$$R\mathbb{E}_{\mathcal{F}_{t-1}}\left[a_t a_t^\top; A_t\right] R^\top = \mathbb{E}_{\mathcal{F}_{t-1}}\left[(Ra_t)(Ra_t)^\top; A_t\right] = \mathbb{E}_{\mathcal{F}_{t-1}}\left[a_t a_t^\top; A_t\right]$$

Now, if M is a symmetric matrix such that $RMR^\top = M$ for all R, it must be that M is a multiple of the identity. Therefore,

$$\mathbb{E}_{\mathcal{F}_{t-1}}\left[a_t a_t^\top; A_t\right] = \alpha I_d.$$

We then get

$$\begin{aligned}\alpha d = \mathrm{Tr}\left(\mathbb{E}_{\mathcal{F}_{t-1}}\left[a_t a_t^\top; A_t\right]\right) &= \mathbb{E}_{\mathcal{F}_{t-1}}\left[\|a_t\|^2; A_t\right] \\ &= \mathbb{E}_{\mathcal{F}_{t-1}}\left[(\|a_t\|^2 - d); A_t\right] + d\mathbb{P}(A_t) = d\mathbb{P}(A_t)\end{aligned}$$

The most important fact for this entire proof, is to notice that the events $B_{t,i}$ are indepedent. This is because, since $q_i \perp q_j$ for $i \neq j$, the $b_t^\top q_i$ and $b_t^\top q_j$ are independent given $\mathcal{F}_{t-1}$. Thus, if $i \neq j$,

$$\begin{aligned}\mathbb{E}_{\mathcal{F}_{t-1}}\left[(Q^\top b_t)(Q^\top b_t)^\top_{i,j}; B_t\right] &= \mathbb{E}_{\mathcal{F}_{t-1}}\left[(q_i^\top b_t)(q_j^\top b_t); B_t\right] \\ &= \mathbb{P}_{\mathcal{F}_{t-1}}\left(\bigcap_{k\neq i,j} B_{t,k}\right) \cdot \mathbb{E}_{\mathcal{F}_{t-1}}\left[q_i^\top b_t; B_{t,i}\right] \cdot \mathbb{E}_{\mathcal{F}_{t-1}}\left[q_j^\top b_t; B_{t,j}\right] \\ &= 0\end{aligned}$$

because if Z is $\mathcal{N}(0,1)$ and $A \in \sigma(Z)$ is a symmetric event, we have $\mathbb{E}[Z; A] = 0$. When $i = j$,

$$\mathbb{E}_{\mathcal{F}_{t-1}}\left[(q_i^\top b_t)^2; B_t\right] = \mathbb{P}_{\mathcal{F}_{t-1}}\left(\bigcap_{k \neq i} B_{t,k}\right) \cdot \mathbb{E}_{\mathcal{F}_{t-1}}\left[(q_i^\top b_t)^2; B_{t,i}\right] = \mathbb{P}_{\mathcal{F}_{t-1}}(B_t)$$

by our judicious choice of $B_{t,i}$ (cf. the lemma before this).

As $Q \in \mathcal{F}_{t-1}$, we thus have

$$\begin{aligned}\mathbb{E}_{\mathcal{F}_{t-1}}\left[Q^\top b_t b_t^\top; B_t\right] &= \mathbb{E}_{\mathcal{F}_{t-1}}\left[(Q^\top b_t)(Q^\top b_t)^\top; B_t\right] \cdot Q^\top \\ &= \mathbb{P}_{\mathcal{F}_{t-1}}(B) I \cdot Q^\top = \mathbb{P}_{\mathcal{F}_{t-1}}(B) Q^\top.\end{aligned}$$

Therefore,

$$\mathbb{E}_{\mathcal{F}_{t-1}}\left[V_t - a_t a_t^\top V_t b_t b_t^\top; A_t, B_t\right] = V_t \mathbb{P}_{\mathcal{F}_{t-1}}(A_t \cap B_t) - \mathbb{E}_{\mathcal{F}_{t-1}}\left[a_t a_t^\top P \Sigma Q^\top b_t b_t^\top; A_t, B_t\right]$$

Now,

$$\begin{aligned}\mathbb{E}_{\mathcal{F}_{t-1}}\left[a_t a_t^\top P\Sigma Q^\top b_t b_t^\top; A_t, B_t\right] &= \mathbb{E}_{\mathcal{F}_{t-1}}\left[a_t a_t^\top; A_t\right] P\Sigma \cdot \mathbb{E}_{\mathcal{F}_{t-1}}\left[Q^\top b_t b_t^\top; B_t\right] \\ &= P\Sigma Q^\top = V_t\end{aligned}$$

Therefore, upon defining

$$Q_t = (V_t - a_t a_t^\top V_t b_t b_t^\top) \mathbb{1}_{A_t \cap B_t}$$

we have

$$\mathbb{E}_{\mathcal{F}_{t-1}}[Q_t] = 0.$$

The next part of the proof is showing that Q_t is $\mathrm{nSG}(C\sqrt{mn}\mathrm{polylog}(dm/\delta)\left\|V\right\|_F)$. First, notice that on B_t,

$$\|b_t\|^2 = \sum_{i=1}^m |q_i^\top b_t| \leq 3m\log(C/\delta)$$

and recall that A_t was defined so that

$$\|a_t\| \leq \sqrt{3d\log(C/\delta)}.$$

So on $A_t \cap B_t$ we have

$$\left\|a_t a_t^\top V_t b_t b_t^\top\right\|_F = |a_t^\top V b_t| \, \|a_t\| \, \|b_t\| \leq |a_t^\top V b_t| \sqrt{9md\log^2(C/\delta)}$$

Our main goal now is to show that $a_t^\top V b_t$ is subgaussian with parameter $O(\mathrm{polylog}(1/\delta)\,\|V\|_F)$. To simplify notation, we will drop the ts and conditioning, it won't be important for the argument. Recall that for $Z \sim \mathcal{N}(0,\sigma^2)$, if $t\sigma^2 < \frac{1}{2}$,

$$\mathbb{E}\left[e^{tZ^2}\right] = (1-2t\sigma^2)^{-1/2}$$

Then given b, since $a^T V b$ has distribution $\mathcal{N}(0, \|Vb\|^2)$, we see immediately that whenever $t\,\|Vb\|^2 < \frac{1}{4}$,

$$\mathbb{E}\left[e^{t(a^T V b)^2} \;\middle|\; b\right] = (1-2t\,\|Vb\|^2)^{-1/2} \le e^{2t\|Vb\|^2}$$

where the latter bound comes from, if $x \le \frac{1}{4}$,

$$\log\left((1-2x)^{-1/2}\right) = -\frac{1}{2}\log(1-2x) \le \frac{1}{2}\cdot\frac{2x}{1-2x} \le 2x$$

While it is now tempting to use $\|Vb\| \le \|V\|_F \cdot \|b\|$, this leads to a bad bound. Instead, observe that Vb is $\mathcal{N}(0, VV^\top)$ distributed.

Now recall that for $Z \sim \mathcal{N}(0, A)$, if $t < 1/(2\lambda_{\max}(A))$,

$$\mathbb{E}\left[e^{t\|Z\|^2}\right] = \det(I - 2tA)^{-1/2}$$

Therefore, as long as $t < 1/(8\sigma_{\max}(V))$, we have that

$$\begin{aligned}\mathbb{E}\left[e^{2t\|Vb\|^2}\right] &= \prod_{i=1}^r (1-4t\sigma_i^2)^{-1/2} = \exp\left(\sum_{i=1}^r -\frac{1}{2}\log\left(1-4t\sigma_i^2\right)\right)\\ &\le \exp\left(4t\sum_{i=1}^r \sigma_r^2\right) = \exp\left(4t\,\|V\|_F^2\right)\end{aligned}$$

We are now almost done. Observe, on the event B,

$$\|Vb\|^2 = b^\top V^\top V b = \sum_{i=1}^r \sigma_i^2 b^\top q_i \le 3\log(C/\delta)\sum_{i=1}^r \sigma_i^2 = 3\log(C/\delta)\,\|V\|_F^2$$

Now, $t\,\|Vb\|^2 \le t3\,\|V\|_F^2\log(C/\delta)$, so choosing $t = 1/(24\,\|V\|_F^2\log(C/\delta))$, we will satisfy all the bounds above ($\sigma_{\max}(V) \le \|V\|_F^2$), and get

$$\mathbb{E}\left[e^{t(a^\top V b)^2}\right] = \mathbb{E}\left[\mathbb{E}\left[e^{t(a^\top V b)^2} \mid b\right]\right] \le e^{4t\|V\|_F^2} \le \exp\left(\frac{1}{6\log(C/\delta)}\right) \le e^{1/6} \le 2$$

Therefore, $a_t^\top V_t b_t$ is $\mathrm{nSG}(\sqrt{24\log(C/\delta)}\,\|V\|_F)$. Combining the previous bounds on $\|a\|$ and $\|b\|$, we get

$$\mathbb{E}_{\mathcal{F}_{t-1}}\left[\exp\left(\frac{\left\|a_t a_t^\top V_t b_t b_t^\top \mathbb{1}_{A_t\cap B_t}\right\|_F^2}{216md\log^3(C/\delta)\,\|V\|_F^2}\right)\right] \le 2$$

Therefore, using $\|Q_t\|_F^2 \leq \|V_t\|^2 + \left\|a_t a_t^\top V_t b_t b_t^\top \mathbb{1}_{A_t \cap B_t}\right\|$, for some absolute constant c',

$$\mathbb{E}_{\mathcal{F}_{t-1}}\left[\exp\left(\frac{\|Q_t\|_F^2}{c' md \log^3(C/\delta)\,\|V\|_F^2}\right)\right] \leq 2$$

This proves that Q_t is nSG($c' \log^{3/2}(C/\delta)\,\|V\|_F$).

Thus, by Lemma 1, for some absolute constant c, with probability at least $1-\delta$,

$$\left\|\sum_{t=0}^{\tau-1} Q_t\right\| \leq c \cdot \theta \sum_{t=0}^{\tau-1} md \log^3(C/\delta)\,\|V_t\|_F^2 + \frac{1}{\theta}\log\frac{2mn}{\delta}.$$

Now, on $\bigcap_{t=0}^{\tau-1} A_t \cap B_t$,

$$\sum_{t=0}^{\tau-1} Q_t = \sum_{t=0}^{\tau-1}(V_t - a_t a_t^\top V_t b_t b_t^\top)$$

As $\mathbb{P}_{\mathcal{F}_{t-1}}(B_{t,i}) \geq 1-\delta$, by union bound $\mathbb{P}_{\mathcal{F}_{t-1}}(B_t) \geq 1 - m\delta$. Further, recall that $\mathbb{P}_{\mathcal{F}_{t-1}}(A_t) \geq 1-\delta$. Therefore,

$$\mathbb{P}_{\mathcal{F}_{t-1}}\left(\bigcap_{t=0}^{\tau-1} A_t \cap B_t\right) \geq 1 - \tau(m+1)\delta$$

so, with probability at least $1 - \delta - \tau(m+1)\delta$,

$$\|E_\tau\| \leq c \cdot \theta \sum_{t=0}^{\tau-1} md \log^3(C/\delta)\,\|V_t\|_F^2 + \frac{1}{\theta}\log\left(\frac{2md}{\delta}\right).$$

After rescaling δ to $\delta/(1+\tau(m+1))$, we have arrived at the desired result. □

3.3.2 Sub-Weibell Random Variables

Definition 3.8. Random variable $X \in \mathbb{R}$ is SW(K, α) for some $K, \alpha > 0$ if

$$\mathbb{P}(|X| \geq s) \leq 2\exp\left(-(s/K)^{1/\alpha}\right) \quad \forall s \geq 0$$

We will need the following lemmas.

Lemma 3.9. *Suppose X and Y are $SW(K_X, \alpha)$ and $SW(K_Y, \alpha)$ respectively. Then XY is $SW(C(K_X \cdot K_Y), 2\alpha)$ and $X+Y$ is $SW(C(K_X+K_Y), \alpha)$ for some absolute constant C.*

Lemma 3.10. *Suppose $X_1, \ldots, X_n$ are identically distributed $SW(K', \alpha)$. Then for some absolute constant $c > 0$, for all $s \geq ncK'$, we have*

$$\mathbb{P}\left(\left|\sum_{i=1}^n X_i\right| \geq s\right) \leq \exp\left(-\left(\frac{s}{ncK'}\right)^{1/\alpha}\right).$$

We will need to frequently bound sums of $\|P_t\|^{2k}$.

Lemma 3.11. *Suppose $a_0, \ldots, a_{\tau-1} \sim \mathcal{N}(0, I_d)$, and $b_0, \ldots, b_{\tau-1} \sim \mathcal{N}(0, I_m)$. Then for any $k > 0$ there exists absolute constants $c, C > 0$ such that for any $0 < \delta < 1$, w.p. $\geq 1 - \delta$,*

$$\sum_{t=0}^{\tau-1} \left\|a_i b_i^\top\right\|_F^{2k} \leq \tau C c^{2k} d^k m^k (1 + \log(1/\delta)^{2k})$$

When $\delta < 1/e$,

$$\sum_{t=0}^{\tau-1} \left\|a_i b_i^\top\right\|_F^{2k} \leq 2\tau C c^{2k} d^k m^k \log(1/\delta)^{2k}.$$

Proof. $(a_i)_k^2$ and $(b_j)_l^2$ are $SW(1,1)$. Thus their product is $SW(c,2)$. Thus $\left\|a_i b_i^T\right\|_F^2 = \sum_{k,l} (a_i)_k^2 (b_j)_k^2$ is $SW(cmd, 2)$ for some absolute constant c. Thus, $\left\|a_i b_i^\top\right\|_F^{2k}$ is $SW(c^k m^k d^k, 2k)$. Therefore by the lemma, for $s \geq \tau C c^k m^k d^k$,

$$\mathbb{P}\left(\sum_{t=0}^{\tau-1} \left\|a_i b_i^\top\right\|_F^{2k} \geq s\right) \leq \exp\left(-\Big(\frac{s}{\tau C c^k d^k m^k}\Big)^{1/2k}\right)$$

the lemma follows by choosing $s = (1 + \log(1/\delta)^{2k}) \tau C c^k d^k m^k$. □

3.4 Function Decrease

First, we prove the following crucial lemma.

Lemma 3.12 (Function decrease for rank r symmetric EGGROLL)**.** *Suppose at each time t, the algorithm performs the update step with rank $r \leq dm$,*

$$X_{t+1} = X_t - \eta g_u^r(X_t)$$

where

$$g_\sigma^r(X_t) := \frac{1}{r} \sum_{i=1}^{r} \frac{f(X_t + \sigma P_{t,i}) - f(X_t - \sigma P_{t,i})}{2\sigma} P_{t,i}$$

where $P_{t,i} = a_{t,i} b_{t,i}^\top$ where each $a_{t,i}, b_{t,i}$ are drawn i.i.d. from $\mathcal{N}(0, I_d)$, and $\mathcal{N}(0, I_m)$ respectively, and $\sigma > 0$ is the smoothing radius.

Then there exists absolute constants $c > 0, C \geq 1$ such that, for any $T \in \mathbb{Z}^+$ and $1 \leq \tau' \leq T$, and $0 < \delta < 1/e$, then upon defining $\mathcal{H}_\tau(\delta)$ to be the event on which the inequality

$$\begin{aligned} f(X_\tau) - f(X_0) \leq &-\frac{\eta}{2} \sum_{t=0}^{\tau-1} \frac{1}{r} \sum_{i=1}^{r} |\langle \nabla f(X_t), P_{t,i} \rangle|^2 + \tau C c^4 \eta \sigma^2 L^2 (dm)^2 \log\left(\frac{T}{\delta}\right)^4 \\ &+ \eta^2 \frac{cdm\chi^4}{r} \sum_{t=0}^{\tau-1} \|\nabla f(X_t)\|^2 + \eta^2 \tau C c^6 \sigma^2 L^3 (dm)^3 \log\left(\frac{T}{\delta}\right)^6 \end{aligned}$$

holds, (where $\chi \coloneqq \log\bigl(Cm^2 drT/\delta\bigr)$), we have

$$\mathbb{P}\left(\bigcap_{\tau=1}^{\tau'} \mathcal{H}_\tau(\delta)\right) \geq 1 - \frac{3\tau'}{T}\delta$$

Proof. For each $t \in \{-1, \ldots, \tau\}$, let

$$\mathcal{F}_t = \sigma(X_0, \{a_{0,i}\}_{i=1}^r, \ldots, \{a_{t,i}\}_{i=1}^r, \{b_{0,i}\}_{i=1}^r, \ldots, \{b_{t,i}\}_{i=1}^r)$$

be the sigma-algebra containing the information available at time t, and as usual write $P_{t,i} = a_{t,i}b_{t,i}^\top$.

Observe that, by mean value theorem, there exists some $s_{t,i} \in [-1, 1]$ such that

$$g_\sigma(X_t) = \frac{1}{r}\sum_{i=1}^r \frac{f(X_t + \sigma P_{t,i}) - f(X_t - \sigma P_{t,i})}{2\sigma} P_{t,i} = \frac{1}{r}\sum_{i=1}^r \nabla f(X_t + \sigma s_{t,i} P_{t,i}) P_{t,i}$$

Then we have that

$$X_{t+1} - X_t = -\eta \cdot \frac{1}{r}\sum_{i=1}^r (\langle \nabla f(X_t), P_{t,i}\rangle P_{t,i} + \langle \nabla f(X_t + \sigma s_{t,i} P_{t,i}) - \nabla f(X_t), P_{t,i}\rangle P_{t,i})$$

$$\begin{aligned} f(X_{t+1}) - f(X_t) &\leq \langle X_{t+1} - X_t, \nabla f(X_t)\rangle + \frac{L}{2}\left\|X_{t+1} - X_t\right\|^2 \\ &\leq -\eta \cdot \frac{1}{r}\sum_{i=1}^r |\langle \nabla f(X_t), P_{t,i}\rangle|^2 \\ &\quad - \eta \cdot \frac{1}{r}\sum_{i=1}^r \langle \nabla f(X_t), P_{t,i}\rangle \langle \nabla f(X_t + \sigma_{t,i} P_{t,i}) - \nabla f(X_t), P_{t,i}\rangle \\ &\quad + \frac{L\eta^2}{2}\left\|\frac{1}{r}\sum_{i=1}^r (\langle \nabla f(X_t), P_{t,i}\rangle P_{t,i} + \langle \nabla f(X_t + \sigma s_{t,i} P_{t,i}) - \nabla f(X_t), P_{t,i}\rangle P_{t,i})\right\|^2 \end{aligned}$$

Now notice, by using $ab \leq (a^2 + b^2)/2$,

$$-\frac{\eta}{r}\sum_{i=1}^r \langle \nabla f(X_t), P_{t,i}\rangle \langle \nabla f(X_t + \sigma_{t,i} P_{t,i}) - \nabla f(X_t), P_{t,i}\rangle \leq \frac{\eta}{2r}\sum_{i=1}^r (|\langle \nabla f(X_t)\rangle, P_{t,i}|^2 + \sigma^2 L^2 \left\|P_{t,i}\right\|^4)$$

Also, by $\left\|a + b\right\|^2 \leq 2\left\|a\right\|^2 + 2\left\|b\right\|^2$,

$$\begin{aligned} &\left\|\frac{1}{r}\sum_{i=1}^r (\langle \nabla f(X_t), P_{t,i}\rangle P_{t,i} + \langle \nabla f(X_t + \sigma s_{t,i} P_{t,i}) - \nabla f(X_t), P_{t,i}\rangle P_{t,i})\right\|^2 \\ &\leq 2\left\|\frac{1}{r}\sum_{i=1}^r \langle \nabla f(X_t), P_{t,i}\rangle P_{t,i}\right\|^2 + \frac{2}{r}\sum_{i=1}^r \left\|\langle \nabla f(X_t + \sigma s_{t,i} P_{t,i}) - \nabla f(X_t), P_{t,i}\rangle P_{t,i})\right\|^2 \\ &\leq 2\left\|\frac{1}{r}\sum_{i=1}^r \langle \nabla f(X_t), P_{t,i}\rangle P_{t,i}\right\|^2 + \frac{2\sigma^2 L^2}{r}\sum_{i=1}^r \left\|P_{t,i}\right\|^6 \end{aligned}$$

Therefore,

$$f(X_{t+1}) - f(X_t) \le -\frac{\eta}{2r}\sum_{i=1}^{r} |\langle \nabla f(X_t), P_{t,i}\rangle|^2 + \frac{\eta\sigma^2 L^2}{2r}\sum_{i=1}^{r} \|P_{t,i}\|^4 + L\eta^2 \left\| \frac{1}{r}\sum_{i=1}^{r}\langle \nabla f(X_t), P_{t,i}\rangle P_{t,i} \right\|^2 + \frac{\eta^2 L^3\sigma^2}{r}\sum_{i=1}^{r}\|P_{t,i}\|^6$$

Now,

$$\left\| \frac{1}{r}\sum_{i=1}^{r}\langle \nabla f(X_t), P_{t,i}\rangle P_{t,i} \right\|^2 \le 2\left\| \frac{1}{r}\sum_{i=1}^{r}(\nabla f(X_t) - \langle \nabla f(X_t), P_{t,i}\rangle P_{t,i}) \right\|^2 + 2\|\nabla f(X_t)\|^2$$

By our proposition, with probability $1-\delta$,

$$\left\| \sum_{i=1}^{r}(\nabla f(X_t) - \langle \nabla f(X_t), P_{t,i}\rangle P_{t,i}) \right\| \le c\cdot \theta dmr \log^3\left(\frac{Crm}{\delta}\right)\|\nabla f(X_t)\|^2 + \frac{1}{\theta}\log\left(\frac{Cm^2dr}{\delta}\right)$$

As $\nabla f(X_t)$ is fixed given $\mathcal{F}_{t-1}$, optimizing over θ, this is at most

$$\sqrt{cdmr}\log\bigl(Cm^2dr/\delta\bigr)^2\|\nabla f(X_t)\|.$$

Therefore for some irrelevant absolute constant c,

$$\left\| \frac{1}{r}\sum_{i=1}^{r}\langle \nabla f(X_t), P_{t,i}\rangle P_{t,i} \right\|^2 \le c\frac{dm}{r}\log\bigl(Cm^2dr/\delta\bigr)^4\|\nabla f(X_t)\|^2 \tag{3.1}$$

By union bound, this will hold jointly for all $0 \le t \le \tau' - 1$ with probability at least $1 - \tau'\delta$.

Now by old lemma, for some absolute constants C, c possibly different from before, with probability at least $1-\delta$,

$$\frac{1}{r}\sum_{t=0}^{\tau-1}\sum_{i=1}^{r}\|P_{t,i}\|^{2k} \le 2\tau C c^{2k}(dm)^k\log(1/\delta)^{2k}.$$

If k is fixed, by union bound, this will hold jointly for all $1 \le \tau \le \tau'$ with probability $1-\tau'\delta$. Therefore, it will hold for both $k = 2$ and 3, for all $1 \le \tau \le \tau'$, with probability at least $1 - 2\tau'\delta$.

Therefore,

$$f(X_\tau) - f(X_0) \le -\frac{\eta}{2r}\sum_{t=0}^{\tau-1}\sum_{i=1}^{r}|\langle \nabla f(X_t), P_{t,i}\rangle|^2 + \tau C c^4\eta\sigma^2L^2(dm)^2\log(1/\delta)^4 + \eta^2\frac{cdm}{r}\log\bigl(Cm^2dr/\delta\bigr)^4\sum_{t=0}^{\tau-1}\|\nabla f(X_t)\|^2 + \eta^2\tau Cc^6\sigma^2L^3(dm)^3\log(1/\delta)^6$$

holds for all $1 \le \tau \le \tau'$ with probability at least $1 - 3\tau\delta$. Equivalently,

$$\mathbb{P}\left(\bigcap_{\tau=1}^{\tau'} \mathcal{H}_\tau(\delta)\right) \ge 1 - 3\tau'\delta$$

Similarly, we see that, as when τ is fixed we need only Equation 3.7 to hold jointly over $0 \le t \le \tau - 1$, while the bound on the perturbation norms hold with probability at least $1 - 2\delta$, so,

$$\mathbb{P}(\mathcal{H}_\tau(\delta)) \ge 1 - (\tau + 2)\delta$$

After rescaling δ to δ/T, we have arrived at the desired result. $\square$

We will need the following Bernstein inequality for sub-exponential random variables (Vershynin, 2018, Theorem 2.8.1).

Lemma 3.13 (Bernstein concentration inequality)**.** *Let $X_1, \ldots, X_n$ be independent, mean-zero, σ-subexponential random variables, i.e.*

$$\mathbb{E}\left[\exp\left(\frac{|X_i|}{\sigma}\right)\right] \le 2 \quad \text{for all } i \in [n].$$

Then there exists an absolute constant c such that for any $s \ge 0$,

$$\mathbb{P}\left(\sum_{i=1}^{n} X_i \ge s\right) \le \exp\left(-c \min\left\{\frac{s^2}{n\sigma^2}, \frac{s}{\sigma}\right\}\right).$$

We then prove the following lemma.

Lemma 3.14. *There exists an absolute constant $c_2 \ge 1$ such that, if*

$$t_f(\delta) = \frac{c_2}{m} \log\left(\frac{T}{\delta}\right), \quad \delta > 0,$$

and

$$\mathcal{B}_{t_0}(\delta; k) = \bigcup_{t=t_0}^{t_0+k-1} \left\{\frac{1}{m} \sum_{i=1}^{m} |a_{t,i}^\top \nabla f(X_t) b_{t,i}|^2 \ge \frac{1}{2} \|\nabla f(X_t)\|^2\right\}$$

we have

$$\mathbb{P}(\mathcal{B}_{t_0}(\delta; k)) \ge 1 - \frac{\delta}{T}$$

for $0 < \delta < 1$, $t_0 \ge 0$ and $k \ge t_f(\delta)$.

Proof. By using the same technique from the proof of Theorem 3, we know that the set of random variables $\left\{\|\nabla f(X_t)\|^2 - |a_{t,i}\nabla f(X_t)b_{t,i}|^2\right\}_{i=1}^m$ are independent, $c\,\|\nabla f(X_t)\|^2$ subexponential for some absolute constant $c > 0$ given $\mathcal{F}_{t-1}$. Now define

$$E_t = \left\{\frac{1}{m}\sum_{i=1}^m |a_{t,i}^\top \nabla f(X_t)b_{t,i}|^2 < \frac{1}{2}\|\nabla f(X_t)\|^2\right\}$$

Therefore,

$$\begin{aligned}
\mathbb{P}_{\mathcal{F}_{t-1}}(E_t) &= \mathbb{P}_{\mathcal{F}_{t-1}}\left(\frac{1}{m}\sum_{i=1}^m |a_{t,i}^\top \nabla f(X_t)b_{t,i}|^2 \geq \|\nabla f(X_t)\|^2\right) \\
&= \mathbb{P}_{\mathcal{F}_{t-1}}\left(\sum_{i=1}^m \left(\|\nabla f(X_t)\|^2 - |a_{t,i}^\top \nabla f(X_t)b_{t,i}|^2\right) > \frac{m}{2}\|\nabla f(X_t)\|^2\right) \\
&\leq \exp\Big(-c' \min\left\{\frac{m}{4c^2}, \frac{m}{2c}\right\}\Big) \leq \exp\big(-c''m\big)
\end{aligned}$$

for some absolute constant c'', where the last line follows by using Bernstein concentration inequality.

Now,

$$\begin{aligned}
\mathbb{P}\left(\bigcap_{t=t_0}^{t_0+k-1} E_t\right) &= \mathbb{P}\left(\bigcap_{t=t_0}^{t_0+k-1}\left\{\frac{1}{m}\sum_{i=1}^m |a_{t,i}^\top \nabla f(X_t)b_{t,i}|^2 < \frac{1}{2}\|\nabla f(X_t)\|^2\right\}\right) \\
&= \mathbb{E}\left[\prod_{t=t_0}^{t_0+k-2} \mathbb{1}_{E_t}\mathbb{E}_{\mathcal{F}_{t_0+k-2}}[\mathbb{1}_{E_{t_0+k-1}}]\right] \leq \cdots \leq \exp\big(c''mk\big)
\end{aligned}$$

therefore by choosing $k \geq t_f(\delta)$ we get

$$\mathbb{P}\left(\bigcap_{t=t_0}^{t_0+k-1}\left\{\frac{1}{m}\sum_{i=1}^m |a_{t,i}\nabla f(X_t)b_{t,i}|^2 < \frac{1}{2}\|\nabla f(X_t)\|^2\right\}\right) \leq \frac{\delta}{T},$$

which completes the proof. $\square$

Lemma 3.15. *Define*

$$\mathcal{A}_t(\delta) := \left\{\|\nabla f(X_{t+1}) - \nabla f(X_t)\| \leq \frac{\|\nabla f(X_t)\|}{8t_f(\delta)} + \eta\sigma L^2 \cdot \frac{1}{r}\sum_{i=1}^r \|P_{t,i}\|^3\right\}$$

where $t_f(\delta)$ is defined above. Then there exists an absolute constant c_2 such that, whenever η satisfies,

$$\eta L c_2 \sqrt{\frac{dm}{r}} \log\left(\frac{CdmrT}{\delta}\right)^2 \leq \frac{1}{8t_f(\delta)}, \tag{3.2}$$

we have

$$\mathbb{P}(\mathcal{A}_t(\delta)) \geq 1 - \frac{\delta}{T}$$

for any $0 < \delta \leq 1/e$ *and* $T \in \mathbb{Z}^+$.

Proof. By L-lipschitz of ∇f,

$$\begin{aligned}
\|\nabla f(X_{t+1}) - \nabla f(X_t)\| &\leq L\,\|X_{t+1} - X_t\| \\
&\leq \eta L \left\| \frac{1}{r}\sum_{i=1}^{r} a_{t,i}a_{t,i}^\top \nabla f(X_t) b_{t,i} b_{t,i}^\top \right\| \\
&\quad + \eta L \left\| \frac{1}{r}\sum_{i=1}^{r} \langle \nabla f(X_t + \sigma s_{t,i} P_{t,i}) - \nabla f(X_t), P_{t,i}\rangle P_{t,i} \right\|
\end{aligned}$$

By L-lipschitz of $\nabla f(X_t)$ and the Cauchy-Schwarz inequality,

$$\left\| \frac{1}{r}\sum_{i=1}^{r} \langle \nabla f(X_t + \sigma s_{t,i} P_{t,i}) - \nabla f(X_t), P_{t,i}\rangle P_{t,i} \right\| \leq L\sigma \cdot \frac{1}{r}\sum_{i=1}^{r} \|P_{t,i}\|^3$$

Now for some absolute constant $c_2 > 0$ by Theorem 3.7, with probability at least $1-\delta/T$,

$$\left\| \frac{1}{r}\sum_{i=1}^{r} a_{t,i}a_{t,i}^\top \nabla f(X_t) b_{t,i} b_{t,i}^\top \right\| \leq c_2 \log(CdmrT/\delta)^2 \sqrt{\frac{dm}{r}}\, \|\nabla f(X_t)\|$$

plugging in our condition on η, we arrive at the statement of the lemma. □

This next very important lemma is the key that will let us lower bound

$$\sum_{t=0}^{\tau-1} \frac{1}{r}\sum_{i=1}^{r} |\langle \nabla f(X_t), P_t\rangle|^2 \geq \tilde{\Omega}\left(\sum_{t=0}^{\tau-1} \|\nabla f(X_t)\|\right)$$

in some sense.

Denote

$$\mathcal{E}(t_1, t_2, \delta) = \bigcap_{t=t_1}^{t_1+t_2-1} \left\{ \|\nabla f(X_t)\| \geq 8 t_f(\delta)\eta\sigma L^2 \cdot \frac{1}{r}\sum_{i=1}^{r} \|P_{t,i}\|^3 \right\}$$

Lemma 3.16. *Let* $0 < \delta \leq 1/e$ *and* $T \in \mathbb{Z}^+$ *satisfying* $T \geq 2t_f(\delta) + 1$. *Consider any positive integer* $t'_f \leq 2t_f(\delta)$, *and any* $t_0 \in \left\{0, \ldots, T - 1 - t'_f\right\}$. *Suppose* η *satisfies the condition above, then on*

$$\mathcal{E}(t_0, t'_f, \delta) \cap \left(\bigcap_{t=t_0}^{t_0+t'_f-1} \mathcal{A}_t(\delta) \right)$$

we have

$$\frac{1}{2}\|\nabla f(X_0)\| \leq \|\nabla f(X_t)\| \leq 2\|\nabla f(X_t)\|$$

for all $t \in \left\{t_0, \ldots, t_0 + t'_f - 1\right\}$.

Proof. By plugging in the definition for $\mathcal{E}_t(\delta)$ into the definition of $\mathcal{A}_t(\delta)$, we see that on the prescribed event,

$$\|\nabla f(X_{t+1}) - \nabla f(X_t)\| \leq \frac{\|\nabla f(X_t)\|}{4t_f(\delta)}$$

therefore

$$\left(1 - \frac{1}{4t_f(\delta)}\right)\|\nabla f(X_t)\| \leq \|\nabla f(X_{t+1})\| \leq \left(1 + \frac{1}{4t_f(\delta)}\right)\|\nabla f(X_t)\|$$

which leads to

$$\left(1 - \frac{1}{4t_f(\delta)}\right)^{t-t_0}\|\nabla f(X_0)\| \leq \|\nabla f(X_t)\| \leq \left(1 + \frac{1}{4t_f(\delta)}\right)^{t-t_0}\|\nabla f(X_0)\|.$$

for all $t \in \left\{t_0, \ldots, t_0 + t'_f - 1\right\}$. Now using $\log(1+x) \leq x$ for $x \geq 0$ and $-\log(1-x) \leq x/(1-x)$ for $0 \leq x < 1$, we know that

$$\left(1 - \frac{1}{4x}\right)^{2x} \geq e^{-2/3} \geq \frac{1}{2}, \quad \left(1 + \frac{1}{4x}\right)^{2x} \leq e^{1/2} \leq 2$$

Now recalling that $t'_f \leq 2t_f(\delta)$, we are done. ☐

Lemma 3.17. *There exists an absolute constant* $c > 0$ *such that for any* $t \in \mathbb{N}$, *the event*

$$\mathcal{G}_t(\delta) := \left\{\frac{1}{r}\sum_{i=1}^{r}\|P_{t,i}\|^3 \leq c(dm)^{3/2}\log(T/\delta)^3\right\}$$

holds with probability at least $1 - \delta/T$ *for* $0 < \delta \leq 1/e$.

Proof. Obvious from Lemma 3.11 and the Cauchy-Schwarz inequality. ☐

Lemma 3.18. *Let* $0 < \delta \leq 1/e$ *and* $T \in \mathbb{Z}^+$ *such that* $T > 2t_f(\delta) + 1$. *Consider any positive integer* $t'_f \leq 2t_f(\delta)$ *and* $t_0 \in \left\{0, \ldots, T - 1 - t'_f\right\}$. *Suppose* η *satifies the condition above. Then on the event*

$$\mathcal{E}^c(t_0, t'_f, \delta) \cap \left(\bigcap_{t=t_0}^{t_0+t'_f-1}\mathcal{A}_t(\delta)\right) \cap \left(\bigcap_{t=t_0}^{t_0+t_f-1}\mathcal{G}_t(\delta)\right)$$

we have

$$\|\nabla f(X_t)\| \leq 16t_f(\delta)\eta\sigma L^2 c(dm)^{3/2}\log(T/\delta)^3$$

for all $t \in \left\{t_0, \ldots, t_0 + t'_f - 1\right\}$.

Proof. Let t' be the first time in $\{t_0, \dots, t_0 + t_f - 1\}$ such that the gradient is dominated, meaning

$$\|\nabla f(X_{t'})\| \leq 8t_f(\delta)\eta L^2 \sigma \frac{1}{r} \sum_{i=1}^{r} \|P_{t,i}\|^3 .$$

By definition of $\mathcal{G}_{t'}(\delta)$, we must have

$$\|\nabla f(X_{t'})\| \leq 8t_f(\delta)\eta L^2 \sigma c (dm)^{3/2} \log(T/\delta)^3$$

For $j < t'$, using the idea from a previous lemma,

$$\|\nabla f(X_j)\| \leq 2\|\nabla f(X_{t'})\| \leq 16 t_f(\delta)\eta L^2 \sigma c (dm)^{3/2} \log(T/\delta)^3$$

On the other hand, for $t \in \left\{t', \dots, t_0 + t'_f - 1\right\}$, by using definition of $\mathcal{A}_t(\delta)$ and $\mathcal{G}_t(\delta)$, we get

$$\begin{aligned}
\|\nabla f(X_{t+1})\| &\leq \left(1 + \frac{1}{8t_f(\delta)}\right) \|\nabla f(X_t)\| + \eta\sigma L^2 c (dm)^{3/2} \log(T/\delta)^3 \\
&= \left(1 - \frac{1}{8t_f(\delta)}\right)^{t+1-t'} \|\nabla f(X_{t'})\| \\
&\quad + \sum_{i=0}^{t-t'} \left(1 - \frac{1}{8t_f(\delta)}\right)^{t-t'-i} \eta\sigma L^2 c (dm)^{3/2} \log(T/\delta)^3 \\
&\leq \left(1 + \frac{1}{8t_f(\delta)}\right)^{t'_f} \|\nabla f(X_t)\| \\
&\quad + 8t_f(\delta)\left(\left(1 + \frac{1}{8t_f(\delta)}\right)^{t'_f} - 1\right) \eta\sigma L^2 c (dm)^{3/2} \log(T/\delta)^3 \\
&\leq e^{1/4} 8 t_f(\delta) \eta\sigma L^2 c (dm)^{3/2} \log(T/\delta)^3 \\
&\quad + 8t_f(\delta)(e^{1/4} - 1)\eta\sigma L^2 c (dm)^{3/2} \log(T/\delta)^3 \\
&\leq 16 t_f(\delta)\eta\sigma L^2 c (dm)^{3/2} \log(T/\delta)^3
\end{aligned}$$

Where we used $t'_f \leq 2t_f(\delta)$ and $(1 + 1/(8x))^{2x} \leq e^{1/4}$ when $x > 0$. This completes the proof. □

We are now almost ready to prove the main theorem. But first, we need the following lemma.

Lemma 3.19 (Function change for large τ)**.** *Let $0 < \delta \leq 1/e$ and let $\tau \geq t_f(\delta)$ be arbitrary. Split $\{0, \dots, \tau - 1\}$ into $K := \tau / t_f(\delta)$ intervals $J_0, \dots, J_{K-1}$ of size $t_f(\delta)$. Let I_1 denote the set of indices where the gradient always dominates the noise term in J_k, i.e.*

$$I_1 := \left\{ k \, : \, \|\nabla f(X_t)\| \geq 8t_f(\delta)\eta\sigma L^2 \frac{1}{r} \sum_{i=1}^{r} \|P_{t,i}\|^3 \;\; \textit{for all } t \in J_k \right\}$$

Suppose we choose η satisfying

$$\eta \leq \frac{1}{8t_f(\delta)} \min\left\{\sqrt{\frac{r}{dm}} \cdot \frac{1}{c_2 L \log(Cdmr\tau/\delta)^2}, \frac{r}{dm} \cdot \frac{1}{8c\chi^4}\right\}$$

Then on the event

$$\mathcal{E}_\tau(\delta) := \mathcal{H}_\tau(\delta) \cap \left(\bigcap_{t=0}^{\tau-1} \mathcal{A}_t(\delta)\right) \cap \left(\bigcap_{t=0}^{\tau-1} \mathcal{G}_t(\delta)\right) \cap \left(\bigcap_{t=0}^{\tau-1} \mathcal{B}_{kt_f(\delta)}(\delta; t_f(\delta))\right)$$

we have the following upper bound for some absolute constants c, C,

$$\begin{aligned} f(X_\tau) - f(X_0) \leq & -\sum_{k \in I_1} \frac{\eta}{4} \min_{t \in J_k} \|\nabla f(X_t)\|^2 + \tau\eta^4 \frac{cdm\chi^4}{r} t_f(\delta)^2 \sigma^2 L^4 (dm)^3 \log(T/\delta)^6 \\ & + \tau C c^4 \eta \sigma^2 L^2 (dm)^2 \log\left(\frac{T}{\delta}\right)^4 + \eta^2 \tau C c^6 \sigma^2 L^3 (dm)^3 \log\left(\frac{T}{\delta}\right)^6. \end{aligned}$$

Furthermore,

$$\mathbb{P}(\mathcal{E}_\tau(\delta)) \geq 1 - \frac{(4\tau + 2)}{T}\delta.$$

Proof. Any interval $J_k = \{t_0, \ldots, t_0 + t_f - 1\}$ belongs to one of the following two cases:

(Case 1) (Gradient dominates noise): This means for every $t \in J_k$,

$$\|\nabla f(X_t)\| \geq 8t_f(\delta)\eta\sigma L^2 \frac{1}{r} \sum_{i=1}^{r} \|P_{t,i}\|^3$$

By the Lemma 3.16 we have

$$\min_{t \in J_k} \|\nabla f(X_t)\| \geq \frac{1}{4} \max_{t \in J_k} \|\nabla f(X_t)\|$$

On the event $\mathcal{B}_{kt_f(\delta)}(\delta; t_f(\delta))$, there exists some $t \in J_k$ such that

$$\frac{1}{r} \sum_{i=1}^{r} |\langle P_{t,i}, \nabla f(X_t)\rangle|^2 \geq \frac{1}{2} \|\nabla f(X_t)\|^2$$

Therefore, using an average,

$$\begin{aligned} \frac{1}{4} \sum_{t \in J_k} \frac{1}{r} \sum_{i=1}^{r} |\langle P_{t,i}, \nabla f(X_t)\rangle|^2 \geq \frac{1}{4} \min_{t \in J_k} \|\nabla f(X_t)\| \geq \frac{1}{64} \max_{t \in J_k} \|\nabla f(X_t)\|^2 \\ \geq \frac{1}{64t_f(\delta)} \sum_{t \in J_k} \|\nabla f(X_t)\|^2 \end{aligned}$$

By choosing

$$\frac{\eta cdm\chi^4}{r} \leq \frac{1}{64t_f(\delta)}$$

we have

$$\begin{aligned}
&-\frac{\eta}{2}\sum_{t\in J_k}\frac{1}{r}\sum_{i=1}^{r}|\langle P_{t,i},\nabla f(X_t)\rangle|^2 + \frac{\eta^2 cdm\chi^4}{r}\sum_{t\in J_k}\|\nabla f(X_t)\|^2 \\
&\leq -\frac{\eta}{4}\sum_{t\in J_k}\frac{1}{r}\sum_{i=1}^{r}|\langle P_{t,i},\nabla f(X_t)\rangle|^2 \leq -\frac{\eta}{4}\min_{t\in J_k}\|\nabla f(X_t)\|^2 .
\end{aligned}$$

(Case 2) (Gradient does not dominate noise): This means that there exists some $t \in J_k$ such that

$$\|\nabla f(X_t)\| \leq 8t_f(\delta)\eta\sigma L^2 \frac{1}{r}\sum_{i=1}^{r}\|P_{t,i}\|^3$$

Then by Lemma 3.18, we know that, for all $t \in J_k$,

$$\|\nabla f(X_t)\| \leq 16t_f(\delta)\eta\sigma L^2 c(dm)^{3/2}\log(T/\delta)^3$$

Every J_k falls into one of the above two cases. Now we can complete the proof.

$$\begin{aligned}
&-\frac{\eta}{2}\sum_{t=0}^{\tau-1}\frac{1}{r}\sum_{i=1}^{r}|\langle P_{t,i},\nabla f(X_t)\rangle|^2 + \eta^2\frac{cdm\chi^4}{r}\sum_{t=0}^{\tau-1}\|\nabla f(X_t)\|^2 \\
&= \sum_{k\in I_1}\left(-\frac{\eta}{2}\sum_{t\in J_k}\frac{1}{r}\sum_{i=1}^{r}|\langle P_{t,i},\nabla f(X_t)\rangle|^2 + \eta^2\frac{cdm\chi^4}{r}\sum_{t\in J_k}\|\nabla f(X_t)\|^2\right) \\
&\quad + \sum_{k\in I_1^c}\left(-\frac{\eta}{2}\sum_{t\in J_k}\frac{1}{r}\sum_{i=1}^{r}|\langle P_{t,i},\nabla f(X_t)\rangle|^2 + \eta^2\frac{cdm\chi^4}{r}\sum_{t\in J_k}\|\nabla f(X_t)\|^2\right) \\
&\leq -\sum_{k\in I_1}\frac{\eta}{4}\min_{t\in J_k}\|\nabla f(X_t)\|^2 + \sum_{k\in I_1^c} t_f(\delta)\left(\eta^4\frac{cdm\chi^4}{r}256t_f(\delta)^2\sigma^2L^4c(dm)^3\log(T/\delta)^6\right) \\
&\leq -\sum_{k\in I_1}\frac{\eta}{4}\min_{t\in J_k}\|\nabla f(X_t)\|^2 + \tau\eta^4\frac{cdm\chi^4}{r}256t_f(\delta)^2\sigma^2L^4c(dm)^3\log(T/\delta)^6
\end{aligned}$$

and thus by the definition of $\mathcal{H}_\tau(\delta)$ in Lemma 3.12,

$$\begin{aligned}
f(X_\tau) - f(X_0) &\leq -\sum_{k\in I_1}\frac{\eta}{4}\min_{t\in J_k}\|\nabla f(X_t)\|^2 + \tau\eta^4\frac{cdm\chi^4}{r}256t_f(\delta)^2\sigma^2L^4c(dm)^3\log(T/\delta)^6 \\
&\quad + \tau Cc^4\eta\sigma^2L^2(dm)^2\log\left(\frac{T}{\delta}\right)^4 + \eta^2\tau Cc^6\sigma^2L^3(dm)^3\log\left(\frac{T}{\delta}\right)^6 .
\end{aligned}$$

Furthermore,

$$\begin{aligned}\mathbb{P}(\mathcal{E}_\tau(\delta)^c) &\leq \mathbb{P}(\mathcal{H}_\tau(\delta)^c) + \sum_{t=0}^{\tau-1}\mathbb{P}(\mathcal{A}_t(\delta)^c) + \sum_{t=0}^{\tau-1}\mathbb{P}(\mathcal{G}_t(\delta)^c) + \sum_{k=0}^{K-1}\mathbb{P}\Big(\mathcal{B}_{kt_f(\delta)}(\delta; t_f(\delta))^c\Big)\\ &\leq \frac{(\tau+2)}{T}\delta + \frac{\tau}{T}\delta + \frac{\tau}{T}\delta + \frac{K}{T}\delta \leq \frac{(4\tau+2)}{T}\delta\end{aligned}$$

which completes the proof. □

Theorem 3.20 (Symmetric EGGROLL Convergence)**.** *Let $0 < \delta < 1/e$ be arbitrary. Suppose we choose σ, η and T such that*

$$\begin{aligned}\sigma^2 &\leq \min\left\{\frac{\varepsilon^2}{1024Cc^4L^2(dm)^2\log(T/\delta)^4}, \frac{\varepsilon^2}{8096rL^3(dm)^2\log(T/\delta)^2}\right\}\\ \eta &\leq \frac{1}{8t_f(\delta)}\min\left\{\sqrt{\frac{r}{dm}}\cdot\frac{1}{cL\log(Cdmr\tau/\delta)^2}, \frac{r}{dm}\cdot\frac{1}{8c\chi^4}\right\},\\ T &\geq \max\left\{4, \frac{512t_f(\delta)(f(X_0)-f^*)}{\eta\varepsilon^2}\right\}.\end{aligned}$$

Then with probability at least $1-5\delta$ there are at most $T/4$ iterations for which $\|\nabla f(X_t)\| \geq \varepsilon$.

Proof. Without loss of generality assume that $t_f(\delta)$ divides T and split $\{0,\dots,T-1\}$ into $K = T/t_f(\delta)$ intervals $J_0,\dots,J_{K-1}$. Let I_1 denote the set of indices k such that for every $t \in J_k$,

$$\|\nabla f(X_t)\| \geq 8t_f(\delta)\eta\sigma L^2\frac{1}{r}\sum_{i=1}^{r}\|P_{t,i}\|^3.$$

Denote

$$\mathcal{E}_T(\delta) \coloneqq \mathcal{H}_T(\delta) \cap \left(\bigcap_{t=0}^{T-1}\mathcal{A}_t(\delta)\right) \cap \left(\bigcap_{t=0}^{T-1}\mathcal{G}_t(\delta)\right) \cap \left(\bigcap_{k=0}^{K-1}\mathcal{B}_{kt_f(\delta)}(\delta; t_f(\delta))\right)$$

For the rest of the proof assume we are working on $\mathcal{E}_T(\delta)$. By Lemma 3.19, and our choices of η and δ in the statement of the theorem, we have

$$\begin{aligned}f(X_T) - f(X_0) \leq &-\sum_{k\in I_1}\frac{\eta}{4}\min_{t\in J_k}\|\nabla f(X_t)\|^2 + T\eta^4\frac{cdm\chi^4}{r}t_f(\delta)^2\sigma^2L^4(dm)^3\log(T/\delta)^6\\ &+ TCc^4\eta\sigma^2L^2(dm)^2\log\left(\frac{T}{\delta}\right)^4 + \eta^2TCc^6\sigma^2L^3(dm)^3\log\left(\frac{T}{\delta}\right)^6\end{aligned}$$

Assume that there are more than $T/4$ iterations with $\|\nabla f(X_t)\| \geq \varepsilon$. Let I_ε denote the set of indices k where there is a $t \in J_k$ with $\|\nabla f(X_t)\| \geq \varepsilon$. Thus by pigeonhole principle, $|I_\varepsilon| \geq T/4t_f(\delta)$. By our choices of parameters, it can be shown that

$$16t_f(\delta)\eta\sigma L^2c(dm)^{3/2}\log(T/\delta)^3 < \varepsilon$$

While by Lemma 3.18, if $k \in I_1^c$,

$$\|\nabla f(X_t)\| \leq 16 t_f(\delta)\eta\sigma L^2 c(dm)^{3/2}\log(T/\delta)^3$$

Therefore, $I_\varepsilon \subset I_1$. Now by Lemma 3.16, for any $k \in I_1$,

$$\frac{1}{2}\left\|\nabla f(X_{kt_f(\delta)})\right\| \leq \|\nabla f(X_t)\| \leq 2\left\|\nabla f(X_{kt_f(\delta)})\right\|, \quad \forall t \in J_k$$

This implies that for any $k \in I_\varepsilon$, $\min_{t\in J_k}\|\nabla f(X_t)\|^2 \geq \frac{1}{16}\varepsilon^2$, and consequently,

$$-\sum_{k\in I_1}\frac{\eta}{4}\min_{t\in J_k}\|\nabla f(X_t)\|^2 \leq -\sum_{k\in I_\varepsilon}\frac{\eta}{2}\cdot\frac{\varepsilon^2}{16} \leq -\frac{T}{4t_f(\delta)}\cdot\frac{\eta}{4}\cdot\frac{\varepsilon^2}{16} = -\frac{T\eta\varepsilon^2}{256 t_f(\delta)}$$

Hence,

$$\begin{aligned} f(X_T) - f(X_0) \leq & -\frac{T\eta\varepsilon^2}{256 t_f(\delta)} + T\eta^4\frac{cdm\chi^4}{r}t_f(\delta)^2\sigma^2L^4(dm)^3\log(T/\delta)^6 \\ & + TCc^4\eta\sigma^2L^2(dm)^2\log\left(\frac{T}{\delta}\right)^4 + \eta^2TCc^6\sigma^2L^3(dm)^3\log\left(\frac{T}{\delta}\right)^6 \end{aligned}$$

By our choice of parameters,

$$\begin{aligned} & T\eta^4\frac{cdm\chi^4}{r}t_f(\delta)^2\sigma^2L^4(dm)^3\log(T/\delta)^6 + TCc^4\eta\sigma^2L^2(dm)^2\log\left(\frac{T}{\delta}\right)^4 \\ & + \eta^2TCc^6\sigma^2L^3(dm)^3\log\left(\frac{T}{\delta}\right)^6 \leq \frac{T\eta\varepsilon^2}{512t_f(\delta)} \end{aligned}$$

We thus get

$$f(X_T) - f(X_0) \leq -\frac{T\eta\varepsilon^2}{512 t_f(\delta)}$$

Therefore, as long as

$$T \geq \frac{512 t_f(\delta)(f(x_0) - f^*)}{\eta\varepsilon^2}$$

we would get $f(X_T) < f^*$, a contradiction. Thus, we may conclude that on the event $\mathcal{E}_T(\delta)$ there are at most $T/4$ iterations for which $\|\nabla f(X_t)\| \geq \varepsilon$.

Finally,

$$\begin{aligned} \mathbb{P}(\mathcal{E}_T^c) & \leq \mathbb{P}(\mathcal{H}_T^c) + \sum_{t=0}^{T-1}\mathbb{P}(\mathcal{A}_t(\delta)) + \sum_{t=0}^{T-1}\mathbb{P}(\mathcal{G}_t(\delta)) + \sum_{k=0}^{K-1}\mathbb{P}\Big(\mathcal{B}_{kt_f(\delta)}(\delta; t_f(\delta))\Big) \\ & \leq \frac{(T+2)\delta}{T} + \delta + \delta + \frac{K\delta}{T} \leq 5\delta. \end{aligned}$$

□

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